Self similar diffusion

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1 The self-similar framework

1.1 Rescaling into a self-similar frame

Let $u_i^*(\mathbf{x}^*, t^*)$, $i = 1, \ldots, d$, satisfy an evolution equation of the form

$$\frac{\partial u_i^*}{\partial t^*} = R_i^*(u^*), \tag{1}$$

on $\mathbb{R}^d$. We rescale amplitude, length, and time by

$$u_i = \frac{u_i^*}{U}, \qquad x_j = \frac{x_j^*}{L}, \qquad t = \frac{t^*}{s(t)}, \tag{2}$$

where the amplitude scale U and the length scale L depend on the non dimensional time t. The timescale is determined by specifiying the clock

$$\tau = \frac{\mathrm{d}t^*}{\mathrm{d}t}. \tag{3}$$

The clock τ is problem specific and can contain the time dependent length and amplitude scales. It is defined in §4.1 and §5.1 for the two cases considered here.

Throughout, a prime denotes differentiation with respect to t and we use the following abbreviations

$$\sigma_L = \frac{L'}{L}, \qquad \sigma_U = \frac{U'}{U}. \tag{4}$$

Differentiating $x_j = x_j^*/L$ with respect to t at fixed x_j^* gives, for each component j,

$$\frac{\mathrm{d}x_j}{\mathrm{d}t} = -x_j^* \frac{L'}{L^2} = -x_j\, \sigma_L. \tag{5}$$

Applying the chain rule to the left-hand side of (1), and using (3) and (5),

$$\begin{aligned} \frac{\partial u_i^*}{\partial t^*} &= \frac{\mathrm{d}t}{\mathrm{d}t^*} \left(\frac{\partial (U u_i)}{\partial t} + U \frac{\partial u_i}{\partial x_j} \frac{\mathrm{d}x_j}{\mathrm{d}t} \right) \\ &= \frac{U}{\tau} \left(\frac{\partial u_i}{\partial t} + \sigma_U\, u_i - \sigma_L\, x_j \frac{\partial u_i}{\partial x_j} \right). \end{aligned} \tag{6}$$

Equating (6) to R_i^* and solving for $\partial u_i/\partial t$ gives the result used throughout this document. Under the rescaling (2), the equation (1) becomes

$$\frac{\partial u_i}{\partial t} = \underbrace{\sigma_L\, x_j \frac{\partial u_i}{\partial x_j} - \sigma_U\, u_i}_{\text{scaling terms}} + \underbrace{\frac{\tau}{U}\, R_i^*(Uu,\, Lx)}_{\text{physical terms}}. \tag{7}$$

Accordingly we split (7) into two flows. The *scaling operator* $\Phi_{\Delta t}^{(\mathrm{S})}$ advances the scaling terms,

$$\Phi_{\Delta t}^{(\mathrm{S})}(u_i) = u_i + \int_t^{t+\Delta t} \left(\sigma_L\, x_j \frac{\partial u_i}{\partial x_j} - \sigma_U\, u_i \right) \mathrm{d}t, \tag{8}$$

and the *physics operator* $\Phi^{(\mathrm{P})}_{\Delta t}$ advances the rest,

$$\Phi^{(\mathrm{P})}_{\Delta t}(u_i) = u_i + \int_t^{t+\Delta t} \frac{\tau}{U}\, R_i^*(Uu,\, Lx)\, \mathrm{d}t. \tag{9}$$

Both are written as the field plus the integral of the terms they advance. $\Phi^{(\mathrm{S})}$ is obtained in closed form below and is shared; $\Phi^{(\mathrm{P})}$ is equation-specific and is supplied in §4.2 and §5.6.

1.2 The scaling operator

Evaluating $\Phi^{(\mathrm{S})}_{\Delta t}$ amounts to solving

$$\frac{\partial u_i}{\partial t} = \sigma_L\, x_j \frac{\partial u_i}{\partial x_j} - \sigma_U\, u_i. \tag{10}$$

This is a linear, first-order PDE and can be solved by the method of characteristics. Note that the scaling terms do not couple the components of u_i, so the argument below is componentwise.

Let $x_j(t)$ denote a characteristic curve satisfying

$$\frac{\mathrm{d}x_j}{\mathrm{d}t} = -x_j\, \sigma_L. \tag{11}$$

The total derivative of u_i along such a curve is

$$\frac{\mathrm{d}u_i}{\mathrm{d}t} = \frac{\partial u_i}{\partial t} + \frac{\partial u_i}{\partial x_j}\frac{\mathrm{d}x_j}{\mathrm{d}t} = \frac{\partial u_i}{\partial t} - \sigma_L\, x_j \frac{\partial u_i}{\partial x_j}, \tag{12}$$

so along the characteristic (10) reduces to the ordinary differential equation

$$\frac{\mathrm{d}u_i}{\mathrm{d}t} = -\sigma_U\, u_i. \tag{13}$$

Integrating (11) from t_k to t_{k+1} gives

$$\ln x_j(t_{k+1}) - \ln x_j(t_k) = -\big[\ln L_{k+1} - \ln L_k\big] \quad \Longrightarrow \quad x_j(t_{k+1}) = x_j(t_k)\, \frac{L_k}{L_{k+1}}, \tag{14}$$

so that the characteristic arriving at the evaluation point $x_j \equiv x_j(t_{k+1})$ originated at

$$x_j(t_k) = x_j\, \frac{L_{k+1}}{L_k}. \tag{15}$$

Integrating (13) gives

$$u_i(t_{k+1}) = u_i(t_k)\, U_k/U_{k+1}. \tag{16}$$

Combining this with the characteristic origin (15) and the initial condition $u_i(t_k, \cdot)$ gives the scaling operator in closed form,

$$\Phi^{(\mathrm{S})}_{\Delta t}(u_i)(\mathbf{x}) = \frac{U_k}{U_{k+1}}\, u_i\!\left(\mathbf{x}\, \frac{L_{k+1}}{L_k}\right). \tag{17}$$

Writing the step ratios

$$\lambda_{U,k+1} = \frac{U_{k+1}}{U_k}, \qquad \lambda_{L,k+1} = \frac{L_{k+1}}{L_k}, \tag{18}$$

this is

$$\Phi^{(\mathrm{S})}_{\Delta t}(u_i)(\mathbf{x}) = \frac{1}{\lambda_{U,k+1}}\, u_i(\mathbf{x}\, \lambda_{L,k+1}). \tag{19}$$

1.2.1 The scaling operator in $\boldsymbol{\kappa}$-space

For equations formulated spectrally it is useful to have the transform of the scaling terms. Using $\widehat{\cdot}$ for the spatial Fourier transform,

$$\hat{u}_i(\boldsymbol{\kappa}) = \int u_i(\mathbf{x})\, e^{-\mathrm{i}\kappa_j x_j}\, \mathrm{d}\mathbf{x}, \tag{20}$$

and writing the spread term in conservative form,

$$x_j \frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j}\left(x_j u_i\right) - d\, u_i, \tag{21}$$

gives

$$\widehat{x_j \frac{\partial u_i}{\partial x_j}} = \mathrm{i}\kappa_j \left(\mathrm{i}\frac{\partial \hat{u}_i}{\partial \kappa_j}\right) - d\,\hat{u}_i = -\left(\kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + d\,\hat{u}_i\right), \tag{22}$$

so that the transform of (10) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left(\kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + d\,\hat{u}_i\right) - \sigma_U\, \hat{u}_i. \tag{23}$$

This is again a linear, first-order PDE, now with characteristics in $\boldsymbol{\kappa}$ satisfying

$$\frac{\mathrm{d}\kappa_j}{\mathrm{d}t} = \kappa_j\, \sigma_L, \tag{24}$$

along which (23) reduces to

$$\frac{\mathrm{d}\hat{u}_i}{\mathrm{d}t} = -\left(d\,\sigma_L + \sigma_U\right) \hat{u}_i. \tag{25}$$

Integrating (24) from t_k to t_{k+1} gives $\kappa_j(t_{k+1}) = \kappa_j(t_k)\,\lambda_{L,k+1}$, so the characteristic arriving at $\boldsymbol{\kappa}$ originated at $\boldsymbol{\kappa}/\lambda_{L,k+1}$, while integrating (25) gives $\hat{u}_i(t_{k+1}) = \hat{u}_i(t_k)/(\lambda_{U,k+1}\lambda_{L,k+1}^{d})$. The scaling operator in $\boldsymbol{\kappa}$-space is therefore

$$\widehat{\Phi^{(\mathrm{S})}_{\Delta t}(u_i)}(\boldsymbol{\kappa}) = \frac{1}{\lambda_{U,k+1}\,\lambda_{L,k+1}^{d}}\, \hat{u}_i\!\left(\frac{\boldsymbol{\kappa}}{\lambda_{L,k+1}}\right). \tag{26}$$

This agrees with transforming (19) directly: the substitution $\mathbf{x}' = \mathbf{x}\,\lambda_{L,k+1}$ in (20) contributes the Jacobian $\lambda_{L,k+1}^{-d}$ and maps $\boldsymbol{\kappa}$ to $\boldsymbol{\kappa}/\lambda_{L,k+1}$. A growing length scale ($\lambda_{L,k+1} > 1$) thus compresses u_i toward $\mathbf{x} = 0$ and correspondingly stretches $\hat{u}_i$ toward larger $|\boldsymbol{\kappa}|$.

1.3 Spectral interpolation

The closed form (19) requires u_i at the rescaled points $\mathbf{x}\,\lambda_{L,k+1}$, which are not grid points. On a uniform periodic grid the represented field is band-limited, so it can be evaluated at arbitrary points exactly, with no interpolation error, by summing its truncated Fourier series there:

$$u_i(\mathbf{x}\,\lambda_{L,k+1}) = \sum_{\boldsymbol{\kappa}\in\mathcal{K}} \hat{u}_i(\boldsymbol{\kappa})\, \exp(\mathrm{i}\,\kappa_j x_j\, \lambda_{L,k+1}), \tag{27}$$

where $\mathcal{K}$ is the set of represented wavenumbers and $\hat{u}_i(\boldsymbol{\kappa})$ are the normalized transform coefficients of u_i. This is the defining property of the scheme: the scaling step introduces no spatial discretization error of its own.

Queries outside the box. The true problem we are trying to approximate is posed on an infinite box, with u_i going to zero as it approaches the boundaries. For $\lambda_{L,k+1} > 1$ it is possible that $|x_j\,\lambda_{L,k+1}| > H/2$, in which case we would like the rescaling in (27) to return 0. However, this is not the current behavior: instead, it wraps around the periodic box and returns a potentially nonzero value from the other end of the domain. Therefore, to obtain the desired behavior we modify the scaling step to

$$u_i(\mathbf{x}\,\lambda_{L,k+1}) = \chi(\mathbf{x}\,\lambda_{L,k+1}) \sum_{\boldsymbol{\kappa}\in\mathcal{K}} \hat{u}_i(\boldsymbol{\kappa})\, \exp(\mathrm{i}\,\kappa_j x_j\, \lambda_{L,k+1}), \qquad \chi(\mathbf{y}) = \prod_d \mathbb{1}[\,|y_d| \le H/2\,]. \tag{28}$$

1.4 Constraint functionals

We will set the ratios $\lambda_{U,k+1}, \lambda_{L,k+1}$ that the scaling step (19) depends on by requiring the rescaled field to carry prescribed values of two functionals F_1, F_2 of the form

$$F_m\left[\frac{1}{\lambda_U}\,u(x\lambda_L)\right] = \lambda_U^{-p_m}\,\lambda_L^{-q_m}\,F_m[u], \qquad m = 1, 2. \tag{29}$$

The targets F_1, F_2 are fixed once, from the initial condition (§2.2). Given a field u with current values $F_m^{\text{in}} = F_m[u]$, imposing $\lambda_U^{-p_m}\lambda_L^{-q_m}F_m^{\text{in}} = F_m$ and taking logarithms gives the linear system

$$\begin{bmatrix} p_1 & q_1 \\ p_2 & q_2 \end{bmatrix} \begin{bmatrix} \ln\lambda_U \\ \ln\lambda_L \end{bmatrix} = \begin{bmatrix} \ln F_1^{\text{in}} - \ln F_1 \\ \ln F_2^{\text{in}} - \ln F_2 \end{bmatrix}, \tag{30}$$

which is uniquely solvable when $p_1q_2 - p_2q_1 \neq 0$, i.e. when each functional responds to amplitude and to length rescaling differently from the other, and is solved in closed form. Once F_m^{in} is evaluated, no further evaluation of the scaled field is needed to find the ratios.

In both cases the second functional is a second moment, and we normalize it by the target value of the zeroth moment F_1. As the normalization is a constant, it changes neither the exponents nor the ratios the constraints produce; it gives F_2 the units of a squared length, a mean-square width. The functionals themselves are equation-specific (§4.3, §5.7); the substep that applies (30) is ConstrainedScale (§2.5).

1.5 Power-law ansatz and the step in physical time

The algorithm also needs to *predict* scale ratios over a fraction of a step, at points where no constraint is available to fix them (§2.7). For this we assume that both scales follow a power law in physical time,

$$X = A_X\,(t^* + \alpha_X)^{\beta_X}, \qquad X \in \{U, L\}, \tag{31}$$

with a prefactor $A_X > 0$, an offset α_X, and an exponent β_X. The prefactor cannot be dropped: the scales are normalized to $U_0 = L_0 = 1$ at the initial condition (§2.2), before the flow has reached the self-similar state, so nothing ties that normalization to the one of the power law. The prefactor cancels from the ratio over a fraction i of step k,

$$\frac{X_{k+i}}{X_k} = \left(1 + \frac{i\,\Delta t_k^*}{t_k^* + \alpha_X}\right)^{\beta_X}. \tag{32}$$

Since the time step $\Delta t_k = t_{k+1} - t_k$ is taken in the self-similar time t, the corresponding physical increment must be obtained from the clock,

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau\,\mathrm{d}t, \tag{33}$$

which can be approximated with a three-point Adams–Bashforth-style quadrature:

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau\,\mathrm{d}t \approx w_0\,\tau_k + w_1\,\tau_{k-1} + w_2\,\tau_{k-2}. \tag{34}$$

The coefficients can be computed as follows.

$$\begin{aligned} w_0 &= \frac{h^3/3 + (2h_1 + h_2)\,h^2/2 + h_1(h_1 + h_2)\,h}{h_1(h_1 + h_2)}, \\ w_1 &= -\frac{h^3/3 + (h_1 + h_2)\,h^2/2}{h_1 h_2}, \\ w_2 &= \frac{h^3/3 + h_1\,h^2/2}{h_2(h_1 + h_2)}, \end{aligned} \tag{35}$$

where $h = \Delta t_k$, $h_1 = \Delta t_{k-1}$, $h_2 = \Delta t_{k-2}$. In this work the step is constant, $\Delta t_k = \Delta t$ for every k (§2.3). With $h = h_1 = h_2 = \Delta t$ the weights reduce to those of the third-order Adams–Bashforth method,

$$w_0 = \frac{23}{12}\,\Delta t, \qquad w_1 = -\frac{16}{12}\,\Delta t, \qquad w_2 = \frac{5}{12}\,\Delta t. \tag{36}$$

Eliminating $t_k^* + \alpha_X = (X_k/A_X)^{1/\beta_X}$ from (32) and taking logarithms gives

$$\ln \frac{X_{k+i}}{X_k} = \beta_X \ln\!\left(1 + \exp\!\left(\ln i + \ln \Delta t_k^* - \frac{\ln X_k - \ln A_X}{\beta_X}\right)\right). \tag{37}$$

The algorithm uses this with $\ln A_X = 0$, as the function

$$\ln \frac{X_{k+i}}{X_k} = G_X(i, \beta_X; k), \qquad G_X(i, \beta; k) = \beta \ln\!\left(1 + \exp\!\left(\ln i + \ln \Delta t_k^* - \frac{\ln X_k}{\beta}\right)\right), \tag{38}$$

in both directions: it is evaluated forward, with $i = \frac{1}{2}$, to predict a half-step ratio (§2.5), and inverted for β, with $i = 1$, to estimate the exponents from the realized history (§2.8).

The prefactor within a step. Setting $\ln A_X = 0$ is not an assumption that $A_X = 1$; the step does not need A_X. By (32) the ratio depends on the pair $(\beta_X,\, t_k^* + \alpha_X)$, and expanding in the step,

$$\ln \frac{X_{k+i}}{X_k} = \frac{\beta_X}{t_k^* + \alpha_X}\, i\, \Delta t_k^* \left(1 - \frac{i\, \Delta t_k^*}{2\,(t_k^* + \alpha_X)} + \dots\right). \tag{39}$$

The leading term depends on the pair only through $\beta_X/(t_k^* + \alpha_X)$, which is the local rate σ_X/τ (§3.1). The inversion of §2.8 fixes exactly this combination, by matching the realized ratio of the step just taken, whatever $\ln A_X$ is assumed. With $\ln A_X = 0$ the β that does so is an *effective* exponent $\tilde\beta_X$ that absorbs the prefactor: matching $(\ln X_k - \ln A_X)/\beta_X$ with $\ln X_k/\tilde\beta_X$ gives $\tilde\beta_X \approx \beta_X\,(1 + \ln A_X/\ln X_k)$, which tends to β_X as the scales grow. The prefactor only decides how the fixed $\beta_X/(t_k^* + \alpha_X)$ is split between β_X and $t_k^* + \alpha_X$, which enters the correction in (39); that is smaller than the leading term by $i\,\Delta t_k^*/(t_k^* + \alpha_X) = i\,\sigma_X \Delta t_k/\beta_X$, the order of the splitting error itself. The algorithm therefore carries only the effective exponents; the exponents β_X themselves are recovered separately, from the whole settled history, in §3.2.

Logarithmic arithmetic. U_k and L_k may become extremely large or small, so we store $\ln U_k$ and $\ln L_k$ rather than U_k and L_k, and perform all arithmetic on them in the logarithmic number system, where multiplication and division become addition and subtraction, and where sums and differences are evaluated by

$$\log_b(X+Y) = \log_b(X) + \log_b\!\big(1 + b^{\,y-x}\big), \qquad \log_b(X-Y) = \log_b(X) + \log_b\!\big(1 - b^{\,y-x}\big) \quad (X > Y > 0), \tag{40}$$

with $x = \log_b(X)$, $y = \log_b(Y)$. In particular (34) is evaluated as

$$\ln \Delta t_k^* = \ln\!\big(w_0\, e^{\ln \tau_k} + w_1\, e^{\ln \tau_{k-1}} + w_2\, e^{\ln \tau_{k-2}}\big). \tag{41}$$

2 Algorithm

The integrator below is stated once, for any equation that supplies the three items of §1: a clock τ, a physics operator $\Phi^{(\mathrm{P})}$, and a pair of constraint functionals. Everything else — the splitting, the scale bookkeeping, the exponent estimation — is common.

2.1 Domain, grid, and notation

The spatial domain is periodic with a fixed physical side length H and N grid points per direction, centred at the origin, so that $x_j \in [-H/2, H/2)$ with uniform spacing $\Delta x = H/N$ and grid points

$$x_n = -\frac{H}{2} + n\,\Delta x, \qquad n = 0, 1, \dots, N-1 \tag{42}$$

in each of the d directions. Every integral over the domain below is approximated by the Riemann sum

$$\int f(\mathbf{x})\,\mathrm{d}\mathbf{x} \approx (\Delta x)^d \sum_{\mathbf{n}} f(\mathbf{x_n}) \tag{43}$$

on this grid.

An index range abbreviates the corresponding tuple of stored values, e.g. $\ln \tau_{k-2:k} = (\ln \tau_{k-2}, \ln \tau_{k-1}, \ln \tau_k)$. The run constants H, N, the time step Δt of §2.3, the constraint targets F_1, F_2, and whatever physical parameters the equation carries are visible to every subroutine and are omitted from their signatures. The step is constant, $\Delta t_k = \Delta t$; the subscript is kept where a formula holds for a general step. As set out in §1.5, all arithmetic on the scales is done on $\ln U_k, \ln L_k$.

2.2 Initialization

A run is specified by the initial field u_0, the domain parameters H, N, the time step Δt of §2.3, a time horizon T_{end}, the splitting scheme *scheme* $\in \{\textsc{LieTrotter}, \textsc{Strang}\}$ (§2.6, §2.7), for equations that define one, the switch *project* of the optional projection inside Scale (§2.5), for equations that use it (here only Navier–Stokes), the gain $\gamma > 0$ and initial rates $\sigma_{U,\text{init}}, \sigma_{L,\text{init}}$ of the scale relaxation (§5.8; $\gamma = \varnothing$ for none), and initial guesses $\beta_U^{(0)}, \beta_L^{(0)}$, which only seed the bracket of the first root-find for β_U, β_L (§2.8). From these:

- The integration loop starts at $t_0 = 0$ and runs while $t_k < T_{\text{end}}$.
- The self-similar frame is initialized at the initial condition itself, $L_0 = U_0 = 1$, i.e. $\ln L_0 = \ln U_0 = 0$.
- The *target* values F_1, F_2 of the constraint functionals (§1.4) are fixed once, from u_0 itself,

$$F_1 = F_1[u_0], \qquad F_2 = F_2[u_0]. \tag{44}$$

- The flag β_{ok}, which records whether the current β_U, β_L are usable by Strang, starts as $\beta_{\text{ok}} = \text{false}$.
- The misses (134) of u_0 are $\mu_{U,0} = \mu_{L,0} = 0$ by (44). Under scale relaxation the integrated misses start at $I_{X,-1} = T_\mu \sigma_{X,\text{init}}/\gamma$, $X \in \{U, L\}$, so that the scales start out at the rates $\sigma_{X,\text{init}}$ (§5.8).

2.3 Time step

Every step is taken at the same step $\Delta t_k = \Delta t$, a run parameter, so that $t_k = k\,\Delta t$ and the quadrature (34) uses the constant weights (36). A constant step also keeps the samples of every recorded quantity equally spaced in t, which simplifies the time averages of Appendix C.

Nothing adapts the step to the field. When the physics operator has a state-dependent stability limit, Δt must be chosen small enough for the field the run settles into, and whether it is is monitored rather than enforced. Let $\Delta t_{\text{CFL}}(u)$ be the largest step at which Evolve from u stays within the stability limit of its time integrator; for Navier–Stokes it is the advective bound (121). The *realized* CFL number of step k,

$$\text{CFL}_k = \frac{\Delta t}{\Delta t_{\text{CFL}}(u_{k+1})}, \tag{45}$$

measures the step against the field it produced. It is recorded every step, and the run warns the first time it exceeds 1; the remedy is a smaller Δt. The 1D diffusion equation has no such limit, since its propagator (66) is exact at any step.

The end of the run. The loop runs while $t_k < T_{\text{end}}$, and the last step is *not* shortened to land on T_{end}, which would break the constant step. The run overshoots T_{end} by less than one step, which is immaterial on the logarithmic clock t. For the same reason, output taken at prescribed times t is taken at the first step reaching each time, never by shortening a step.

2.4 Top-level integrator

Algorithm 1 Top-level integrator

Require: u_0, H, N, Δt, T_{end}, *scheme*, γ, $\sigma_{U,\text{init}}$, $\sigma_{L,\text{init}}$, $\beta_U^{(0)}$, $\beta_L^{(0)}$

$t_0 \leftarrow 0$, $k \leftarrow 0$
$\ln U_0 \leftarrow 0$, $\ln L_0 \leftarrow 0$
Compute constraint targets F_1, F_2 from u_0 via (44)
$\beta_U \leftarrow \beta_U^{(0)}$, $\beta_L \leftarrow \beta_L^{(0)}$, $\beta_{\text{ok}} \leftarrow$ false
$\mu_{U,0} \leftarrow 0$, $\mu_{L,0} \leftarrow 0$ ▷ (134)
$I_{X,-1} \leftarrow T_\mu \sigma_{X,\text{init}}/\gamma$, $X \in \{U, L\}$ ▷ under scale relaxation, §5.8
Store $(u_0, \ln U_0, \ln L_0, \ln \tau_0, \beta_U, \beta_L, \beta_{\text{ok}}, \mu_{U,0}, \mu_{L,0})$
while $t_k < T_{\text{end}}$ **do**
  if $\gamma \neq \varnothing$ **then**
    $I_{X,k} \leftarrow I_{X,k-1} + \mu_{X,k}\,\Delta t$, $X \in \{U, L\}$ ▷ (144)
    $C \leftarrow (I_{U,k}, I_{L,k})$ ▷ §5.8
  else
    $C \leftarrow \varnothing$
  end if
  if *scheme* = STRANG **and** β_{ok} **then**
    $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{STRANG}(u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L, C)$ ▷ §2.7
  else
    $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{LIETROTTER}(u_k, \ln U_k, \ln L_k, \Delta t, C)$ ▷ §2.6
  end if
  $u_{k+1} \leftarrow \text{APPLYBC}(u_{k+1})$ ▷ equation-specific, §2.5
  $\ln \tau_{k+1} \leftarrow \ln \tau(\ln U_{k+1}, \ln L_{k+1})$ ▷ (63), (84)
  $t_{k+1} \leftarrow t_k + \Delta t$
  if $k + 1 \geq 3$ **then**
    $(\beta_U, \beta_L, \beta_{\text{ok}}) \leftarrow \text{ESTIMATEBETA}(\ln U_{k:k+1}, \ln L_{k:k+1}, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L)$ ▷ §2.8
  end if
  $\mu_{X,k+1} \leftarrow \mu_X[u_{k+1}]$, $X \in \{U, L\}$ ▷ (134)
  Store $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}, \ln \tau_{k+1}, t_{k+1}, \text{CFL}_k, \beta_U, \beta_L, \beta_{\text{ok}}, I_{U,k}, I_{L,k}, \mu_{U,k+1}, \mu_{L,k+1})$ ▷ (45)
  $k \leftarrow k + 1$
end while

The run parameter *scheme* selects the step function. STRANG needs usable estimates of β_U, β_L, so the loop falls back to LIETROTTER whenever β_{ok} is false. This is always the case for the first three steps $k = 0, 1, 2$: the first β update needs four stored states, so it first runs at the end of step $k = 2$. It is also the case for every step that follows a failed β update (§2.8), until a later update succeeds. The β update itself runs regardless of *scheme*; under *scheme* = LIETROTTER the estimates are diagnostic only. Scale relaxation needs no β and so acts from the first step $k = 0$. Without it ($C = \varnothing$) the integrals $I_{X,k}$ are not formed, and the misses $\mu_{X,k}$ are recorded as a diagnostic only.

2.5 Substeps

Both splitting schemes are assembled from the substeps below. None of them applies the boundary treatment, which is handled only by the top-level loop.

Evolve$(u, \Delta t_k)$**.** Applies the physics operator $\Phi^{(\text{P})}_{\Delta t_k}$ of (9): it advances the non-scaling terms of (7) over Δt_k, leaving the scales $\ln U, \ln L$ untouched.

Scale$(u, \ln \lambda_U, \ln \lambda_L)$**.** Applies the scaling operator (19) with prescribed ratios λ_U, λ_L, returning $\lambda_U^{-1} u(\mathbf{x_n}\lambda_L)$ at each grid point, evaluated with the spectral interpolation of §1.3 and set to zero wherever $\mathbf{x_n}\lambda_L$ leaves

the box (28). Here, and wherever the substeps are used, λ_U and λ_L denote the ratio of U, respectively L, after the substep to its value before it. Inside a Lie–Trotter step this is $\lambda_{U,k+1} = U_{k+1}/U_k$ of (18); inside a Strang step it is a ratio across a half step, $U_{k+1/2}/U_k$ or $U_{k+1}/U_{k+1/2}$ (§2.7).

Optional projection. With the run parameter *project* enabled, Scale ends by applying an equation-specific projection $\mathcal{P}$ onto the admissible fields, returning $\mathcal{P}\big(\lambda_U^{-1} u(\mathbf{x_n}\lambda_L)\big)$; with it disabled, or for an equation that defines no $\mathcal{P}$, the projection is the identity. The continuous operator (19) maps a divergence-free field to a divergence-free field, but its evaluation on the grid by interpolation and the out-of-box cutoff does not preserve this exactly. For Navier–Stokes, $\mathcal{P}$ is the Leray projection (250), which restores $\nabla\cdot\tilde{\mathbf{u}} = 0$ (§A.2); the 1D diffusion equation defines none. Because every scaling step runs through Scale, the projection follows the scaling step of LieTrotter and *both* scaling half steps of Strang. Inside ConstrainedScale it is applied after the ratios have been solved, so the returned field meets the targets F_1, F_2 only up to the part the projection removes. The miss does not accumulate: the next ConstrainedScale measures the functionals of the projected field afresh and absorbs it into the next ratios.

ConstrainedScale$(u, \ln U, \ln L, C, \ln\lambda^{\text{pred}} = (0,0))$**.** Rescales u so that the result carries the fixed targets F_1, F_2 of (44) — or, under scale relaxation, so that each log-scale relaxes toward its target over the step — and advances the log-scale state accordingly. Here $\ln\lambda^{\text{pred}} = (\ln\lambda_U^{\text{pred}}, \ln\lambda_L^{\text{pred}})$ are the log-ratios already applied in the step by a predicted half step:

1. **Functionals of the input,** $F_m^{\text{in}} = F_m[u]$, evaluated on the grid via (43).
2. **Ratios** $\ln\lambda_U, \ln\lambda_L$ from the linear system (30), in closed form. This works on the scalars F_m^{in} alone and does not touch the field.
3. **Relaxation.** If a state $C = (I_{U,k}, I_{L,k})$ is given, replace the ratios by (146) (§5.8), in which the ratios of the previous item are the misses μ_X^{P}; with $C = \varnothing$ they are kept.
4. **Rescale,** $u' = \text{Scale}(u, \ln\lambda_U, \ln\lambda_L)$.
5. **Return** u' together with the advanced state $\ln U + \ln\lambda_U$, $\ln L + \ln\lambda_L$.

Without scale relaxation no time increment enters ConstrainedScale: the ratios are fixed by the functionals of the field it is given, not by the elapsed time. The same substep therefore serves as the full scaling step of Lie–Trotter and as the closing scaling half step of Strang; under scale relaxation the two differ only in the log-ratios $\ln\lambda^{\text{pred}}$ already applied by the predicted half step.

PredictScale$(i, \ln U_k, \ln L_k, \ln\tau_{k-2:k}, \Delta t, \beta_U, \beta_L)$**.** Predicts the ratios over the fraction i of step k from the power-law form (31) rather than from the constraints. It evaluates $\ln\Delta t_k^*$ from (41), with the constant-step weights (36), and returns $\big(G_U(i,\beta_U;k),\, G_L(i,\beta_L;k)\big)$ using (38). The same function G_X is inverted for β in §2.8. Both use $\ln A_X = 0$, which the step does not need (§1.5).

ApplyBC(u)**.** Imposes whatever boundary treatment the equation requires, after the full step has been assembled. The 1D diffusion equation clamps its two edge grid points to zero (§4.5). For the Navier–Stokes equations ApplyBC is the identity: the far-field condition is left to the sponge forcing of §5.3, which acts inside Evolve. ApplyBC performs no projection; that is done only inside Scale.

2.6 LieTrotter: Lie–Trotter splitting

A Lie–Trotter step composes a full physics step with a full scaling step, $u_{k+1} = \Phi^{(\text{S})}_{\Delta t_k}\big(\Phi^{(\text{P})}_{\Delta t_k}(u_k)\big)$, with the scaling ratios fixed by the constraints, or under scale relaxation by (146) (§5.8). It uses no estimate of β_U, β_L, so it is valid from the first step. It is written for a general input state $(u, \ln U, \ln L)$ and step Δt_k rather than only a stored state $(u_k, \ln U_k, \ln L_k)$, so that Strang can call it on a half-stepped state.

Algorithm 2 Lie–Trotter step

function LieTrotter($u, \ln U, \ln L, \Delta t_k, C, \ln \lambda^{\text{pred}} = (0,0)$)
  $u^{(\mathrm{P})} \leftarrow$ Evolve($u, \Delta t_k$) ▷ §2.5
  return ConstrainedScale($u^{(\mathrm{P})}, \ln U, \ln L, C, \ln \lambda^{\text{pred}}$) ▷ §2.5
end function

2.7 Strang: Strang splitting

A Strang step is the symmetric composition

$$u_{k+1} = \Phi^{(\mathrm{S})}_{\Delta t_k/2}\Big(\Phi^{(\mathrm{P})}_{\Delta t_k}\big(\Phi^{(\mathrm{S})}_{\Delta t_k/2}(u_k)\big)\Big) \tag{46}$$

of a scaling half step, a full physics step, and a second scaling half step. The last two of these are exactly a Lie–Trotter step — a full physics step followed by a scaling step whose ratios come from the constraints — so Strang is a predicted scaling half step followed by a call to LieTrotter.

Algorithm 3 Strang step

function Strang($u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L, C$)
  $\big(\ln(U_{k+1/2}/U_k),\, \ln(L_{k+1/2}/L_k)\big) \leftarrow$ PredictScale($\frac{1}{2}, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t, \beta_U, \beta_L$) ▷ (47)
  $u_{k+1/2} \leftarrow$ Scale($u_k, \ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k)$)
  $\ln U_{k+1/2} \leftarrow \ln U_k + \ln(U_{k+1/2}/U_k)$; $\ln L_{k+1/2} \leftarrow \ln L_k + \ln(L_{k+1/2}/L_k)$
  $\ln \lambda^{\text{pred}} \leftarrow \big(\ln(U_{k+1/2}/U_k),\, \ln(L_{k+1/2}/L_k)\big)$ ▷ used only if $C \neq \varnothing$, §5.8
  return LieTrotter($u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2}, \Delta t_k, C, \ln \lambda^{\text{pred}}$) ▷ §2.6
end function

No constraint fixes the state at $t_k + \Delta t_k/2$, so the first scaling half step takes its ratios from the current power-law estimates, via PredictScale with $i = 1/2$:

$$\ln \frac{U_{k+1/2}}{U_k} = G_U\big(\tfrac{1}{2}, \beta_U; k\big)\,, \qquad \ln \frac{L_{k+1/2}}{L_k} = G_L\big(\tfrac{1}{2}, \beta_L; k\big)\,. \tag{47}$$

The call to LieTrotter then advances $u_{k+1/2}$ over the full Δt_k, and its ConstrainedScale substep — the second scaling half step — sets $U_{k+1}/U_{k+1/2}$ and $L_{k+1}/L_{k+1/2}$ so that u_{k+1} satisfies the constraints. Under scale relaxation the predicted half step is unchanged and the relaxation governs the *full* step: the closing half step completes the relaxed increment (145) of each log-scale, less the log-ratio $\ln \lambda_X^{\text{pred}}$ the predicted half step already applied (§5.8). The half-step state $(u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2})$ is internal to the step and is not stored. The top-level loop calls Strang only when β_{ok} is true (§2.4), which guarantees that β_U, β_L were fitted to the stored history at the end of the previous step and that $\ln \tau_{k-2}$ exists.

2.8 EstimateBeta: estimating β_U and β_L

After every step with $k+1 \geq 3$, regardless of *scheme*, β_U and β_L are re-estimated from the log-scale history. Four stored states are needed: $\ln X_k$ and $\ln X_{k+1}$ for the ratio, and $\ln \tau_{k-2:k}$ for Δt_k^*. The update inverts the function G_X of (38) with $i = 1$, the same function that PredictScale evaluates forward with $i = 1/2$:

$$G_U(1, \beta_U; k) = \ln U_{k+1} - \ln U_k, \tag{48}$$

and identically with $U \to L$ for β_L. The Δt_k^* inverted here is the one PredictScale used for that step's half step, so prediction and estimation are exact inverses on the same step. Since G_X carries $\ln A_X = 0$, the roots are the effective exponents $\tilde{\beta}_X$ of §1.5, which absorb the prefactor and tend to β_X as the scales grow; we write them β_X in the algorithm.

The sensitivity of β_X to an error in Δt_k^* is weak. Taking logarithms of (38) for a small step, $\ln(\ln X_{k+1} - \ln X_k) \approx \ln \beta_X + \ln \Delta t_k^* - \ln X_k/\beta_X$, so a relative error η in Δt_k^* moves β_X by $\delta\beta_X \approx \beta_X^2 \eta / \ln X_k$, damped as the scales grow.

Each of (48) and its L-counterpart is a single scalar equation in a single unknown (β_U, resp. β_L), solved by `brentq`; the implementer will need to choose a bracketing interval (e.g. centred on the previous β_U, β_L estimate — on $\beta_U^{(0)}, \beta_L^{(0)}$ for the first update — widened adaptively if it fails to bracket a root).

If both root-finds succeed, EstimateBeta returns the new β_U, β_L with $\beta_{\text{ok}} = \text{true}$. If either fails to bracket a root even after widening, the failed estimate keeps its previous value (which seeds the next bracket), and EstimateBeta returns $\beta_{\text{ok}} = \text{false}$, so the next step falls back to LieTrotter. Before $k+1 \geq 3$ no update is made: β_U, β_L stay at $\beta_U^{(0)}, \beta_L^{(0)}$ and β_{ok} stays false.

3 Post-processing

A completed run leaves behind the log-scale history $\ln U_k, \ln L_k$ and the clock history $\ln \tau_k$. From these we recover the exponents β_X of the ansatz (31) and the rate c_τ at which the clock grows, the constants that the budgets of §6 take from a run. This is the offline, global counterpart of EstimateBeta (§2.8), which fits an effective exponent one step at a time with $\ln A_X = 0$; here the whole settled record is used at once, and the exponent is obtained as a ratio of mean rates, in which the prefactor does not enter.

Write

$$\mathcal{L}_k = \ln L_k, \qquad \mathcal{U}_k = \ln U_k, \tag{49}$$

and $\mathcal{X}_k = \ln X_k$ for either scale, $X \in \{U, L\}$, together with the per-step log-ratios $\ln \lambda_{X,k} = \ln X_k - \ln X_{k-1}$ of (18), which for a Strang step combine both half-step ratios. The stored scales are their cumulative sums,

$$\mathcal{L}_k = \mathcal{L}_0 + \sum_{i=1}^{k} \ln \lambda_{L,i}, \qquad \mathcal{U}_k = \mathcal{U}_0 + \sum_{i=1}^{k} \ln \lambda_{U,i}, \tag{50}$$

which grow only linearly in k; unlike L_k and U_k themselves, $\mathcal{L}_k$ and $\mathcal{U}_k$ never overflow.

The fitted range. The rates settle only once the initial transient has passed. We discard the record before a cutoff time t_{cut}, the same discard as in step 1 of §C.1, and fit over the steps $k_{\text{c}} \leq k \leq K$, where $t_{k_{\text{c}}}$ is the first sample time at or after t_{cut} and K is the last step. The fit is made against the self-similar time t, so no physical time need be accumulated.

3.1 The scaling rates under the ansatz

Differentiating the ansatz (31) with respect to t, and using the clock (3) to convert from t^*, the logarithmic rates (4) are

$$\sigma_X = \frac{1}{X}\frac{\mathrm{d}X}{\mathrm{d}t^*}\frac{\mathrm{d}t^*}{\mathrm{d}t} = \frac{\beta_X\,\tau}{t^* + \alpha_X}, \qquad X \in \{U, L\}, \tag{51}$$

independent of the prefactor. The rates are estimated directly from the log-ratios already in hand, without any finite difference of L_k or U_k themselves,

$$\sigma_{X,k} \approx \frac{\ln \lambda_{X,k+1}}{\Delta t_k}, \qquad X \in \{U, L\}. \tag{52}$$

The rates, not the log-ratios $\ln \lambda_{X,k+1}$ themselves, are also what the run's diagnostics plot, since a ratio scales with the step and the rates do not, so runs at different Δt compare directly.

Constant rates. By (51) both rates are constant exactly when the offsets agree, $\alpha_U = \alpha_L = \alpha$, and the clock grows linearly in physical time,

$$\tau = c_\tau\,(t^* + \alpha). \tag{53}$$

The rates are then

$$\sigma_X = c_\tau\,\beta_X, \qquad \frac{\mathrm{d}\ln\tau}{\mathrm{d}t} = \frac{\tau}{t^* + \alpha} = c_\tau, \tag{54}$$

so that c_τ is also the rate at which $\ln\tau$ grows in t. Since the clock is built from the scales, (53) constrains the exponents: the Navier–Stokes clock $\tau = L/U$ requires $\beta_L - \beta_U = 1$, with $c_\tau = A_L/A_U$ (§6.1.3), and the diffusion clock $\tau = L^2/D$ requires $\beta_L = \frac{1}{2}$, with $c_\tau = A_L^2/D$.

3.2 Recovering c_τ and β_X

On a run whose rates have settled, (54) integrates to straight lines in t,

$$\mathcal{X}_k = a_X + c_\tau \beta_X \, t_k, \qquad \ln \tau_k = a_\tau + c_\tau \, t_k, \tag{55}$$

with intercepts a_X, a_τ that depend on the prefactors and the offset. We fit each record $Y \in \{\mathcal{U}, \mathcal{L}, \ln \tau\}$ by least squares against t over the fitted range,

$$s_Y = \frac{\sum_k (t_k - \bar{t})(Y_k - \bar{Y})}{\sum_k (t_k - \bar{t})^2}, \qquad a_Y = \bar{Y} - s_Y \, \bar{t}, \tag{56}$$

where the sums and the means $\bar{t}$, $\bar{Y}$ run over $k_\mathrm{c} \le k \le K$, and, writing $s_X = s_{\mathcal{X}}$ and $s_\tau = s_{\ln \tau}$, read off

$$c_\tau = s_\tau, \qquad \beta_X = \frac{s_X}{s_\tau}, \qquad X \in \{U, L\}. \tag{57}$$

The slope of each line is the mean rate over the fitted range, so c_τ and β_X are averages over the whole settled record. Neither the physical time, the offset nor the prefactors enter, and β_X, a ratio of rates, is free of the relative bias $\ln A_X / \ln X_k$ of the effective exponent of EstimateBeta (§1.5). The prefactors and the offset are not recovered: the budgets of §6 need only the rates, which they take as (156) with c_τ and β_X from (57). In an ensemble (§C.3) each run is fitted on its own record, and its budgets use its own constants.

The clock is a fixed linear combination of the log-scales, $\ln \tau_k = \mathcal{L}_k - \mathcal{U}_k$ (84) or $\ln \tau_k = 2\mathcal{L}_k - \ln D$ (63), and the least-squares slope is linear in the data, so s_τ is the same combination of s_U and s_L. The constraint that (53) places on the exponents therefore holds identically, $\beta_L - \beta_U = 1$ or $\beta_L = \frac{1}{2}$, and is no check on the run. The check is the scatter of each record about its line, the root-mean-square deviation

$$r_Y = \left(\frac{1}{K - k_\mathrm{c} + 1} \sum_{k=k_\mathrm{c}}^{K} (Y_k - a_Y - s_Y \, t_k)^2 \right)^{1/2}. \tag{58}$$

On a settled run the deviation $Y_k - a_Y - s_Y t_k$ fluctuates about zero without a trend; a range that includes the initial transient, before the frame has locked on, bends it. The fit presupposes constant rates rather than testing the ansatz (31): it does not recover the offsets, so it cannot check that $\alpha_U = \alpha_L$, and the steadiness of the record rests on this scatter. Under the scale relaxation of §5.8 the stored $\mathcal{X}_k$ are the relaxed scales, and it is these that are fitted.

4 The 1D diffusion equation

The first instantiation of the framework is the one-dimensional diffusion equation, $d = 1$, with a single scalar component; we drop the component index throughout this section. It serves as a testbed: every operator the framework needs is available in closed form, so the only approximation in a run is the operator splitting itself.

4.1 Governing equation and clock

The dimensional equation is

$$\frac{\partial u^*}{\partial t^*} = D \frac{\partial^2 u^*}{\partial (x^*)^2}, \tag{59}$$

so that, in the notation of (1),

$$R^*(u^*) = D \frac{\partial^2 u^*}{\partial (x^*)^2}. \tag{60}$$

Under the rescaling (2) this term becomes

$$D \frac{\partial^2 (uU)}{\partial (x^*)^2} = DU \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{1}{L} \right) \frac{1}{L} = \frac{DU}{L^2} \frac{\partial^2 u}{\partial x^2}, \tag{61}$$

having used $\mathrm{d}x/\mathrm{d}x^* = 1/L$. The clock is then chosen to make the physical term of the lemma (7) carry unit coefficient,

$$\frac{\tau}{U}\cdot\frac{DU}{L^2} = 1 \quad\Longrightarrow\quad \tau = \frac{L^2}{D}, \tag{62}$$

which is the diffusive time scale of the current length scale, as expected. In log form, as required by (41),

$$\ln\tau_k = 2\ln L_k - \ln D. \tag{63}$$

Substituting (62) into the rescaling lemma (7) gives the equation the algorithm integrates,

$$\frac{\partial u}{\partial t} = \sigma_L\, x\frac{\partial u}{\partial x} - \sigma_U\, u + \frac{\partial^2 u}{\partial x^2}. \tag{64}$$

4.2 The physics step

Because the diffusion term in (64) has unit coefficient, the physics operator (9),

$$\Phi^{(\mathrm{P})}_{\Delta t}(u_k) = u_k + \int_{t_k}^{t_k+\Delta t}\frac{\partial^2 u}{\partial x^2}\,\mathrm{d}t, \tag{65}$$

has the closed-form spectral solution

$$\hat{u}^{(\mathrm{P})}_j = \hat{u}_j\, e^{-\kappa_j^2\Delta t}, \qquad \kappa_j = \frac{2\pi j}{H}, \quad j = 0,\ldots,\lfloor N/2\rfloor, \tag{66}$$

where $\hat{u}_j = \mathrm{RFFT}(u)_j$. This is an exact one-shot spectral propagator, not an approximate time-marching scheme, so no sub-stepping or implicit solve is needed, and the accuracy requirement on Evolve (§2.5) is met trivially at any order. Inverse-transforming gives the advanced field $u^{(\mathrm{P})}(x_n)$.

4.3 Moment constraints

The constraint functionals of §1.4 are taken to be the zeroth and the normalized second moment of the field,

$$F_1[u] = \int u\,\mathrm{d}x \equiv M_0, \qquad F_2[u] = \frac{1}{M_0}\int x^2 u\,\mathrm{d}x \equiv M_2, \tag{67}$$

where the M_0 in F_2 is the target value (44), a constant, not the zeroth moment of the field the functional is applied to. Thus M_0 measures the amount of the conserved quantity and M_2, a mean-square width, its spread. Writing $M_0^{\mathrm{in}}, M_2^{\mathrm{in}}$ for the moments of the field being scaled, and substituting $y = x\,\lambda_L$ in each integral, the scaled field carries

$$\int \frac{1}{\lambda_U}u(x\,\lambda_L)\,\mathrm{d}x = \frac{1}{\lambda_U\lambda_L}\,M_0^{\mathrm{in}}, \tag{68}$$

$$\int x^2\,\frac{1}{\lambda_U}u(x\,\lambda_L)\,\mathrm{d}x = \frac{1}{\lambda_U\lambda_L^3}\,M_2^{\mathrm{in}}, \tag{69}$$

so the homogeneity exponents of (29) are

$$(p_1,q_1) = (1,1), \qquad (p_2,q_2) = (1,3), \tag{70}$$

with $p_1q_2 - p_2q_1 = 2 \neq 0$, so the constraint system (30) is uniquely solvable: the two moments do respond independently to amplitude and to length rescaling. Solving it,

$$\lambda_L = \sqrt{\frac{M_0\,M_2^{\mathrm{in}}}{M_0^{\mathrm{in}}\,M_2}}, \tag{71}$$

$$\lambda_U = \frac{M_0^{\mathrm{in}}}{M_0\,\lambda_L} = \left(\frac{M_0^{\mathrm{in}}}{M_0}\right)^{3/2}\sqrt{\frac{M_2}{M_2^{\mathrm{in}}}}. \tag{72}$$

Because the normalization of F_2 is a constant, these ratios are the same as for the raw second moment $\int x^2 u\,\mathrm{d}x$; written in terms of the raw moments, M_0 cancels from M_2^{in}/M_2.

4.4 Spectral interpolation on the real transform

For $d = 1$ and a real-valued field, the generic interpolation formula (27) is realized on the non-negative frequency components of a real Fast Fourier Transform (RFFT):

$$\Phi^{(\mathrm{S})}_{\Delta t}(u_k)(x) = \frac{1}{\lambda_{U,k+1}}\Bigg\{\hat{u}^{(k)}_0 + 2\sum_{j=1}^{N/2-1}(-1)^j\Bigg[\mathrm{Re}\big(\hat{u}^{(k)}_j\big)\cos\left(\frac{2\pi j\lambda_{L,k+1}x}{H}\right) - \mathrm{Im}\big(\hat{u}^{(k)}_j\big)\sin\left(\frac{2\pi j\lambda_{L,k+1}x}{H}\right)\Bigg] + (-1)^{N/2}\,\hat{u}^{(k)}_{N/2}\cos\left(\frac{\pi N\lambda_{L,k+1}x}{H}\right)\Bigg\}, \tag{73}$$

where $\hat{u}^{(k)}_j = \frac{1}{N}\mathrm{RFFT}(u_k)_j$ for $j = 0, 1, \ldots, N/2$, and the Nyquist term at $j = N/2$ is included only if N is even. The factor of 2 accounts for the Hermitian conjugate modes omitted by the real transform. (73) holds for $|\lambda_{L,k+1}x| \le H/2$; outside that the cutoff (28) sets $\Phi^{(\mathrm{S})}_{\Delta t}(u_k)(x) = 0$.

The alternating sign $(-1)^j$ is the phase-origin correction, and it is not optional. The transform phases its coefficients to the sample index n, whose origin $n = 0$ sits at $x = -H/2$ on the centred grid (42) rather than at $x = 0$. Since the scaling operator (19) dilates about the centre of the domain, each mode must be evaluated at $\lambda_{L,k+1}x + H/2$ — dilate about $x = 0$, then shift to the transform's phase origin — and not at $\lambda_{L,k+1}(x + H/2)$, which dilates the shift along with the coordinate and agrees with it only at $\lambda_{L,k+1} = 1$. As $H/2$ is half a period, that shift contributes the constant phase $\exp(\mathrm{i}\kappa_j H/2) = (-1)^j$. Dropping it altogether returns $u_k(\lambda_{L,k+1}x - H/2)$, the wanted profile displaced by half the domain, rather than the $u_k(\lambda_{L,k+1}x)$ that (19) calls for. The same factor carries to each axis in three dimensions (§A.2).

4.5 Boundary treatment

The substep ApplyBC of §2.5 imposes a homogeneous Dirichlet condition by clamping the two edge grid points of the centred domain (42) after each completed step,

$$u_{k+1}(x_0) \leftarrow 0, \qquad u_{k+1}(x_{N-1}) \leftarrow 0. \tag{74}$$

On the centred grid $x \in [-H/2, H/2)$ these are two distinct grid points, not periodic images of one another.

Open. This sits in tension with the rest of the discretization: the spectral propagator (66) presumes a periodic field, whereas (74) imposes a Dirichlet condition. The spectral interpolation (73) no longer does at the edges: with the cutoff (28) it treats everything beyond the box as zero, which is consistent with the clamp, but it is still a periodic interpolant inside the box. The two are consistent only so long as u is negligible near the domain edges, which requires the initial condition to be localized well inside the domain and the run to be stopped before the profile spreads to the boundary. The scaling step works against this — it is what keeps the profile from spreading in the self-similar frame — but the condition is an assumption of the scheme rather than a consequence of it, and is not currently monitored.

5 The Navier–Stokes equations

The second instantiation is the incompressible Navier–Stokes equations, with $d = 3$ and u_i the velocity field. The initial condition and the three-dimensional spectral interpolation used by Scale are given in Appendix A.

5.1 Governing equations and clock

The equations are solved as a large-eddy simulation (LES). Let $\mathcal{G}_{\Delta^*}$ denote a spatial filter of width Δ^*. A tilde marks a quantity of the filtered (resolved) field, and the same symbol without a tilde the corresponding quantity of the full field: $\tilde{u}^*_i = \mathcal{G}_{\Delta^*}(u^*_i)$ and $\tilde{p}^*$ are the filtered velocity and pressure, while u^*_i and p^* are the full ones. For a quantity derived from the velocity, the tilde means it is evaluated from $\tilde{u}^*_i$, not that it

is filtered; thus $\frac{1}{2}\tilde{u}_i^*\tilde{u}_i^*$ is not $\mathcal{G}_{\Delta^*}(\frac{1}{2}u_i^*u_i^*)$. The effect of the unresolved scales enters through the kinematic subgrid-scale stress

$$\tau_{ij}^* = \mathcal{G}_{\Delta^*}(u_i^*u_j^*) - \tilde{u}_i^*\tilde{u}_j^*, \tag{75}$$

which, like the eddy viscosity below, exists only in the LES and carries no tilde. The dimensional filtered equations are

$$\frac{\partial \tilde{u}_i^*}{\partial t^*} + \tilde{u}_j^*\frac{\partial \tilde{u}_i^*}{\partial x_j^*} = -\frac{1}{\rho}\frac{\partial \tilde{p}^*}{\partial x_i^*} + \nu^*\frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}, \tag{76}$$

together with the continuity equation,

$$\frac{\partial \tilde{u}_i^*}{\partial x_i^*} = 0, \tag{77}$$

so that, in the notation of (1),

$$R_i^*(\tilde{u}^*) = -\tilde{u}_j^*\frac{\partial \tilde{u}_i^*}{\partial x_j^*} - \frac{1}{\rho}\frac{\partial \tilde{p}^*}{\partial x_i^*} + \nu^*\frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}. \tag{78}$$

In addition to the rescaling (2) of $\tilde{u}_i$ and x_j, the pressure is nondimensionalized by the dynamic pressure,

$$\tilde{p} = \frac{\tilde{p}^*}{\rho U^2}, \tag{79}$$

and the subgrid-scale stress, which carries units of velocity squared, by U^2,

$$\tau_{ij} = \frac{\tau_{ij}^*}{U^2}. \tag{80}$$

With the filter width nondimensionalized as $\Delta = \Delta^*/L$, the definitions carry over unchanged, $\tilde{u}_i = \mathcal{G}_\Delta(u_i)$ and $\tau_{ij} = \mathcal{G}_\Delta(u_iu_j) - \tilde{u}_i\tilde{u}_j$. Term by term, the convective, pressure, and subgrid-scale terms scale as U^2/L,

$$\tilde{u}_j^*\frac{\partial \tilde{u}_i^*}{\partial x_j^*} = \frac{U^2}{L}\tilde{u}_j\frac{\partial \tilde{u}_i}{\partial x_j}, \qquad \frac{1}{\rho}\frac{\partial \tilde{p}^*}{\partial x_i^*} = \frac{U^2}{L}\frac{\partial \tilde{p}}{\partial x_i}, \qquad \frac{\partial \tau_{ij}^*}{\partial x_j^*} = \frac{U^2}{L}\frac{\partial \tau_{ij}}{\partial x_j}, \tag{81}$$

while the viscous term scales as ν^*U/L^2,

$$\nu^*\frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} = \frac{\nu^* U}{L^2}\frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j}. \tag{82}$$

The clock is chosen as the inverse of the advection timescale

$$\tau = \frac{L}{U}. \tag{83}$$

In log form, as required by (41),

$$\ln \tau_k = \ln L_k - \ln U_k. \tag{84}$$

Applying $\tau/U = L/U^2$ to (78) and introducing the instantaneous Reynolds number

$$Re = \frac{UL}{\nu^*}, \tag{85}$$

the rescaling lemma (7) gives the nondimensional momentum equation

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L\, x_j\frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U\, \tilde{u}_i - \tilde{u}_j\frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{1}{Re}\frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j}, \tag{86}$$

and the continuity equation is unchanged by the rescaling,

$$\frac{\partial \tilde{u}_i}{\partial x_i} = 0. \tag{87}$$

5.2 Subgrid stress model

The subgrid-scale stress τ^*_{ij} of (76) is closed with an eddy-viscosity model. Its deviatoric part is taken proportional to the resolved strain rate,

$$\tau^*_{ij} - \frac{1}{3}\tau^*_{kk}\delta_{ij} = -2\nu^*_{\mathrm{t}}\tilde{S}^*_{ij}, \qquad \tilde{S}^*_{ij} = \frac{1}{2}\left(\frac{\partial \tilde{u}^*_i}{\partial x^*_j} + \frac{\partial \tilde{u}^*_j}{\partial x^*_i}\right), \tag{88}$$

with the isotropic part $\frac{1}{3}\tau^*_{kk}\delta_{ij}$ absorbed into the pressure. The eddy viscosity ν^*_{t} carries the units of UL, so it is nondimensionalized as $\nu_{\mathrm{t}} = \nu^*_{\mathrm{t}}/(UL)$. Noting that $\tilde{S}^*_{ij} = (U/L)\,\tilde{S}_{ij}$, the nondimensional subgrid stress (80) is

$$\tau_{ij} - \frac{1}{3}\tau_{kk}\delta_{ij} = -2\nu_{\mathrm{t}}\tilde{S}_{ij}. \tag{89}$$

The models below differ only in the choice of ν^*_{t}.

5.2.1 Smagorinsky model

The standard Smagorinsky model takes the eddy viscosity

$$\nu^*_{\mathrm{t}} = (C_s\Delta^*)^2|\tilde{S}^*|, \qquad |\tilde{S}^*| = \sqrt{2\tilde{S}^*_{ij}\tilde{S}^*_{ij}}, \tag{90}$$

where C_s is the Smagorinsky constant and Δ^* is the filter width. With the nondimensional filter width $\Delta = \Delta^*/L$ taken equal to the grid spacing, $\Delta = \Delta x$, the eddy viscosity becomes

$$\nu^*_{\mathrm{t}} = UL\,(C_s\Delta)^2|\tilde{S}|, \qquad \nu_{\mathrm{t}} = (C_s\Delta)^2|\tilde{S}|. \tag{91}$$

5.2.2 Self-similar constant eddy viscosity

The second model replaces the local eddy viscosity by a constant that is uniform in space, chosen so that the dissipative scale of the flow tracks the grid. Throughout this paper the dissipation rate ϵ^* is half the trace of the dissipation tensor (§6.1.2), $\epsilon^* = \nu^*\,\overline{(\partial u^*_i/\partial x^*_k)(\partial u^*_i/\partial x^*_k)}$ (152), the sink in the equation for $\overline{k^*} = \frac{1}{2}\overline{u^*_i u^*_i}$, with ν^*_{t} in place of ν^* in the LES. It is not the thermodynamic dissipation rate $2\nu^*\,\overline{S^*_{ij}S^*_{ij}}$, the rate at which kinetic energy is converted to heat: the two differ by a divergence that vanishes in homogeneous turbulence, where they are equal, but not in the inhomogeneous flow considered here. With zero molecular viscosity, the Kolmogorov scale built from the eddy viscosity and this ϵ^*,

$$\eta^* = \left(\frac{(\nu^*_{\mathrm{t}})^3}{\epsilon^*}\right)^{1/4}, \tag{92}$$

is required to be proportional to the filter width,

$$\eta^* = C_\eta\Delta^*, \tag{93}$$

with C_η a constant. In homogeneous turbulence (92) is the conventional Kolmogorov scale. The dissipation rate is estimated from the large scales of the self-similar frame,

$$\epsilon^* \approx C_\epsilon\frac{U^3}{L}, \tag{94}$$

with C_ϵ a constant, so that (92) and (93) give

$$C_\eta\Delta^* = \left(\frac{(\nu^*_{\mathrm{t}})^3 L}{C_\epsilon U^3}\right)^{1/4}. \tag{95}$$

Solving for ν^*_{t} and writing $\Delta^* = L\Delta$,

$$\nu^*_{\mathrm{t}} = C_\nu(\Delta^*)^{4/3}\frac{U}{L^{1/3}} = C_\nu\Delta^{4/3}\,UL, \qquad \nu_{\mathrm{t}} = C_\nu\Delta^{4/3}, \tag{96}$$

where

$$C_\nu = C_\eta^{4/3} C_\epsilon^{1/3}, \tag{97}$$

and $\Delta = \Delta^*/L = \Delta x$ as in the Smagorinsky model. The eddy viscosity is uniform in space, so with (77) the subgrid stress term reduces to a Laplacian,

$$-\frac{\partial \tau_{ij}^*}{\partial x_j^*} = \nu_\mathrm{t}^* \frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*}, \qquad -\frac{\partial \tau_{ij}}{\partial x_j} = C_\nu \Delta^{4/3} \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j}. \tag{98}$$

In the self-similar frame the model is therefore a constant viscosity, of the same form as the molecular term of (86) with an effective Reynolds number $Re_\mathrm{t} = 1/(C_\nu \Delta^{4/3})$. Unlike Re, which changes with UL as the flow evolves, Re_t is fixed for the run, while in physical variables, where $\Delta^* = L\Delta$ grows with the flow, ν_t^* follows the product UL.

The momentum equation

In the limit of infinite Reynolds number, $1/Re = 0$, and with either model the momentum equation (86) becomes

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L \, x_j \frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U \, \tilde{u}_i - \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[2\nu_\mathrm{t} \, \tilde{S}_{ij} \right], \tag{99}$$

where $\tilde{p}$ now denotes the modified pressure that includes $\frac{1}{3}\tau_{kk}$, and ν_t is given by (91) or (96). For the self-similar constant eddy viscosity the last term is $C_\nu \Delta^{4/3} \, \partial^2 \tilde{u}_i / \partial x_j \partial x_j$, by (98).

5.3 Sponge forcing

The true boundary conditions we would like to enforce is $\tilde{u}_i = 0$ at the edge of the domain. However, we are using a pseudo-spectral solver so this boundary condition cannot be directly enforced. To solve this we introduce a fringe region at the edge of the domain [1], which introduces a forcing term to the right hand side of the Navier–Stokes equation

$$f_i = -W(r) \, \tilde{u}_i, \tag{100}$$

where $r = \sqrt{x_i x_i}$ is the distance from the origin and $W(r)$ is a weighting function that increases with increasing r.

The weighting function is

$$W(r) = W_\mathrm{max} \, S\!\left(\frac{r - r_0}{r_1 - r_0} \right), \tag{101}$$

where the smooth step S is

$$S(\xi) = \begin{cases} 0, & \xi \le 0, \\ \dfrac{1}{1 + \exp\!\left(\dfrac{1}{\xi - 1} + \dfrac{1}{\xi} \right)}, & 0 < \xi < 1, \\ 1, & \xi \ge 1. \end{cases} \tag{102}$$

The function S is C^∞ on the whole real line. Every derivative of it vanishes at $\xi = 0$ and at $\xi = 1$, it rises monotonically between them, and it is antisymmetric about its midpoint, $S(1-\xi) = 1 - S(\xi)$, so that $S(1/2) = 1/2$. The weight (101) is therefore exactly zero in the sphere $r \le r_0$, rises smoothly across the shell $r_0 < r < r_1$, and is exactly W_max for $r \ge r_1$. Adding (100) to (99), the momentum equation actually solved is

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L \, x_j \frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U \, \tilde{u}_i - \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[2\nu_\mathrm{t} \, \tilde{S}_{ij} \right] - W(r) \, \tilde{u}_i. \tag{103}$$

5.4 Fourier representation

Using the Fourier convention (20), the pressure gradient becomes

$$\widehat{\frac{\partial \tilde{p}}{\partial x_i}} = \mathrm{i}\kappa_i \hat{\tilde{p}}. \tag{104}$$

The convective term

$$\hat{N}_i = \widehat{\tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j}}, \tag{105}$$

will be evaluated pseudo-spectrally. How the subgrid stress term

$$\hat{G}_i = \frac{\partial}{\partial x_j} \widehat{\left[2\nu_\mathrm{t}\tilde{S}_{ij}\right]} \tag{106}$$

is evaluated depends on the model of §5.2. In both cases the strain rate is available spectrally,

$$\hat{\tilde{S}}_{ij} = \frac{\mathrm{i}}{2}\left(\kappa_j \hat{\tilde{u}}_i + \kappa_i \hat{\tilde{u}}_j\right). \tag{107}$$

Smagorinsky model. The eddy viscosity (91) depends on the field, so $\hat{G}_i$ is evaluated pseudo-spectrally: $\tilde{S}_{ij}$ is transformed to physical space, the products $|\tilde{S}|$, $\nu_\mathrm{t} = (C_s\Delta)^2|\tilde{S}|$ and $T_{ij} = 2\nu_\mathrm{t}\tilde{S}_{ij}$ are formed there, and the divergence is taken spectrally,

$$\hat{G}_i = \mathrm{i}\kappa_j \hat{T}_{ij}. \tag{108}$$

Since $|\tilde{S}|$ is not a polynomial in $\tilde{u}$, $\hat{G}_i$ carries aliasing that the 2/3 rule does not remove entirely.

Self-similar constant eddy viscosity. The eddy viscosity (96) is uniform, and by (98)

$$\hat{G}_i = -C_\nu \Delta^{4/3}\kappa^2 \hat{\tilde{u}}_i. \tag{109}$$

This is exact and diagonal in $\boldsymbol{\kappa}$: no product is formed in physical space, so $\hat{G}_i$ needs no transforms, carries no aliasing, and stays in the retained band. By (112) it is also solenoidal, $\kappa_i\hat{G}_i = 0$, so it contributes nothing to the pressure below and passes through the projection (118) unchanged.

Sponge forcing. The sponge forcing (100) is formed as a product in physical space and transformed,

$$\hat{F}_i = \widehat{W\tilde{u}_i}, \tag{110}$$

with W tabulated once on the grid (42). Because W is smooth but not band-limited, $\hat{F}_i$ also carries content beyond the retained band, which the 2/3 mask removes along with that of $\hat{N}_i$ and, under the Smagorinsky model, $\hat{G}_i$. The scaling terms transform as in (23) with $d = 3$, so the transform of (103) is

$$\frac{\partial \hat{\tilde{u}}_i}{\partial t} = -\sigma_L\left(\kappa_j \frac{\partial \hat{\tilde{u}}_i}{\partial \kappa_j} + 3\,\hat{\tilde{u}}_i\right) - \sigma_U\,\hat{\tilde{u}}_i - \hat{N}_i - \mathrm{i}\kappa_i\hat{\tilde{p}} + \hat{G}_i - \hat{F}_i, \tag{111}$$

and the transform of the continuity equation (87) is

$$\kappa_i \hat{\tilde{u}}_i = 0. \tag{112}$$

5.5 Pressure and the solenoidal projection

Contracting (111) with κ_i and using the continuity constraint gives

$$0 = -\sigma_L\left(\kappa_i\kappa_j \frac{\partial \hat{\tilde{u}}_i}{\partial \kappa_j} + d\,\kappa_i\hat{\tilde{u}}_i\right) - \sigma_U\,\kappa_i\hat{\tilde{u}}_i - \kappa_i\hat{N}_i - \mathrm{i}\kappa^2\hat{\tilde{p}} + \kappa_i\hat{G}_i - \kappa_i\hat{F}_i. \tag{113}$$

The remaining scaling contribution also vanishes: differentiating (112) with respect to κ_j gives

$$\frac{\partial}{\partial\kappa_j}\left(\kappa_i\hat{\tilde{u}}_i\right)=\hat{\tilde{u}}_j+\kappa_i\frac{\partial\hat{\tilde{u}}_i}{\partial\kappa_j}=0\quad\Longrightarrow\quad\kappa_i\frac{\partial\hat{\tilde{u}}_i}{\partial\kappa_j}=-\hat{\tilde{u}}_j,\tag{114}$$

so that

$$\kappa_i\kappa_j\partial\hat{\tilde{u}}_i/\partial\kappa_j=-\kappa_j\hat{\tilde{u}}_j=0.\tag{115}$$

Therefore, the equation for pressure is

$$\mathrm{i}\kappa^2\hat{\tilde{p}}=-\kappa_i\left(\hat{N}_i-\hat{G}_i+\hat{F}_i\right)\quad\Longrightarrow\quad\hat{\tilde{p}}=\mathrm{i}\,\frac{\kappa_l\left(\hat{N}_l-\hat{G}_l+\hat{F}_l\right)}{\kappa^2},\qquad\kappa\neq0,\tag{116}$$

with $\hat{\tilde{p}}$ at $\kappa=0$ fixed arbitrarily, since only its gradient enters the momentum equation. Substituting (116) into (111), the pressure term becomes

$$-\mathrm{i}\kappa_i\hat{\tilde{p}}=\frac{\kappa_i\kappa_l}{\kappa^2}\left(\hat{N}_l-\hat{G}_l+\hat{F}_l\right),\tag{117}$$

which combines with $-\hat{N}_i+\hat{G}_i-\hat{F}_i$ to give the solenoidal projection of the convective, subgrid stress, and sponge terms together. Defining the projection tensor

$$P_{il}(\boldsymbol{\kappa})=\delta_{il}-\frac{\kappa_i\kappa_l}{\kappa^2},\tag{118}$$

the evolution equation in fourier space is (7),

$$\frac{\partial\hat{\tilde{u}}_i}{\partial t}=\underbrace{-\sigma_L\left(\kappa_j\frac{\partial\hat{\tilde{u}}_i}{\partial\kappa_j}+3\,\hat{\tilde{u}}_i\right)-\sigma_U\,\hat{\tilde{u}}_i}_{\text{scaling terms}}\underbrace{-P_{il}(\boldsymbol{\kappa})\left(\hat{N}_l-\hat{G}_l+\hat{F}_l\right)}_{\text{physical terms}}.\tag{119}$$

5.6 The physics step

The physics operator (9) is the flow of the bracketed physical terms of (119),

$$\Phi^{(\mathrm{P})}_{\Delta t}\left(\hat{\tilde{u}}_i(t)\right)=\hat{\tilde{u}}_i(t)-\int_t^{t+\Delta t}P_{il}(\boldsymbol{\kappa})\left(\hat{N}_l-\hat{G}_l+\hat{F}_l\right)\mathrm{d}t.\tag{120}$$

We take $\Phi^{(\mathrm{P})}$ to be a single step of the classical fourth-order Runge–Kutta method (RK4) applied to the full physical right-hand side, the subgrid stress term $\hat{G}_i$ and the sponge forcing $\hat{F}_i$ marched together with the convection. The convective term has eigenvalues on the imaginary axis with $|\lambda|\leq\max|\tilde{\mathbf{u}}|\,\kappa_{\max}$, where $\kappa_{\max}=2\pi(N/3)/H$ is the top mode retained by the 2/3 rule, and RK4 is stable on the imaginary axis for $|\lambda\Delta t|\leq2\sqrt{2}$. This gives the advective limit

$$\Delta t\leq\Delta t_{\mathrm{CFL}}(\tilde{u})=\frac{2\sqrt{2}}{\max|\tilde{\mathbf{u}}|\,\kappa_{\max}},\tag{121}$$

which depends on the state and is monitored through the realized CFL number (45) of §2.3. The sponge forcing and, under the self-similar constant eddy viscosity, the subgrid term are linear decays with eigenvalues on the negative real axis, where RK4 is stable for $|\lambda\Delta t|\lesssim2.785$. This gives the sponge limit $\Delta t\,W_{\max}\leq2.785$ and the diffusive limit $\Delta t\,C_\nu\Delta^{4/3}\kappa_{\max}^2\leq2.785$, both fixed by the run parameters. None of the three is enforced: the constant step Δt must be chosen to satisfy them.

5.7 Constraint functionals

The constraint functionals are built from the resolved kinetic energy, which we define first.

Kinetic energy. The resolved kinetic energy is

$$\tilde{k}^* = U^2\,\tilde{k}, \qquad \tilde{k} = \frac{1}{2}\tilde{u}_i\tilde{u}_i. \tag{122}$$

Contracting (75) gives the filtered kinetic energy of the full field [2],

$$\mathcal{G}_\Delta(k) = \tilde{k} + k_{\text{sgs}}, \qquad k_{\text{sgs}} = \frac{1}{2}\tau_{kk}. \tag{123}$$

The isotropic part of τ_{ij} is absorbed into the pressure, so the solver never computes k_{sgs} and we must estimate it. It enters neither the integration nor the constraints below; we use it only as a diagnostic of how much energy the resolved field misses. For both subgrid stress models we estimate it with the Smagorinsky eddy viscosity (91), so that the estimate is the same function of the resolved field in either case. Equating the production $\nu_{\text{t}}|\tilde{S}|^2 = C_s^2\Delta^2|\tilde{S}|^3$ of subgrid kinetic energy to its sink $C_d k_{\text{sgs}}^{3/2}/\Delta$, with C_d a constant of order one, gives

$$k_{\text{sgs}} = \left(\frac{C_s^2}{C_d}\right)^{2/3}\Delta^2|\tilde{S}|^2, \tag{124}$$

with $k_{\text{sgs}}^* = U^2 k_{\text{sgs}}$.

The functionals. The length and velocity scales will be set such that the zeroth and normalized second radial moments of the resolved energy density $\tilde{u}_i\tilde{u}_i = 2\tilde{k}$, twice the resolved kinetic energy per unit mass (122), are constant. Mathematically, these moments are defined as

$$F_1[\tilde{u}] = \int \tilde{u}_i\tilde{u}_i\,\mathrm{d}\mathbf{x} \equiv E_0, \qquad F_2[\tilde{u}] = \frac{1}{E_0}\int r^2\,\tilde{u}_i\tilde{u}_i\,\mathrm{d}\mathbf{x} \equiv E_2, \tag{125}$$

where $r^2 = x_j x_j$ is the squared distance from the centre of the domain (42), and the E_0 in F_2 is the target value (44), a constant. As in the 1D case, E_0 measures the amplitude of the field and E_2 its spread.

The functionals are built from the resolved field alone: the subgrid kinetic energy k_{sgs} of (123) is neglected, so E_0 is twice the resolved kinetic energy rather than that of the full field, and E_2 is the mean-square radius of the resolved energy. This is accurate to the extent that the resolved field carries most of the energy. We monitor the resolved fraction $K_{\text{res}}/(K_{\text{res}} + K_{\text{sgs}})$, with $K_{\text{res}} = \int \tilde{k}\,\mathrm{d}\mathbf{x}$ and $K_{\text{sgs}} = \int k_{\text{sgs}}\,\mathrm{d}\mathbf{x}$ from the estimate (124), a posteriori against the criterion that a well-resolved LES carries at least 80% of the kinetic energy in the resolved field [2].

Writing $E_0^{\text{in}}, E_2^{\text{in}}$ for the moments of the field being scaled and substituting $\mathbf{y} = \mathbf{x}\,\lambda_L$ in each integral, with $d = 3$, the scaled field carries

$$\int \frac{1}{\lambda_U^2}\,\tilde{u}_i\tilde{u}_i(\mathbf{x}\,\lambda_L)\,\mathrm{d}\mathbf{x} = \frac{1}{\lambda_U^2\lambda_L^3}\,E_0^{\text{in}}, \tag{126}$$

$$\frac{1}{E_0}\int r^2\,\frac{1}{\lambda_U^2}\,\tilde{u}_i\tilde{u}_i(\mathbf{x}\,\lambda_L)\,\mathrm{d}\mathbf{x} = \frac{1}{\lambda_U^2\lambda_L^5}\,E_2^{\text{in}}, \tag{127}$$

so the homogeneity exponents of (29) are

$$(p_1, q_1) = (2, 3), \qquad (p_2, q_2) = (2, 5), \tag{128}$$

with $p_1 q_2 - p_2 q_1 = 4 \neq 0$, so the constraint system (30) is uniquely solvable. Solving it,

$$\lambda_L = \sqrt{\frac{E_0\,E_2^{\text{in}}}{E_0^{\text{in}}\,E_2}}, \tag{129}$$

$$\lambda_U = \sqrt{\frac{E_0^{\text{in}}}{E_0\,\lambda_L^3}}, \tag{130}$$

where the length ratio is identical in form to its 1D counterpart (71).

5.8 Relaxation of the scales

The constrained ratios (129), (130) fix E_0, E_2 exactly at every step, and so pass every fluctuation of the moments of the evolved field into U and L. As a result, the velocity and length scales don't exactly follow the assumed power law distribution. Instead for $X \in \{U, L\}$, we have

$$X = \overline{X} + X', \tag{131}$$

where $\overline{X}$ is the ensemble average of X, which we assume still follows the assumed power law, and X' is a mean-zero fluctuation. When $X' \neq 0$ and we nondimensionalize an ensemble-averaged quantity

$$\overline{Q^*} = \overline{U^{m_U} L^{m_L} Q}, \tag{132}$$

whose units are set by the exponents m_U, m_L, the time-dependent U and L cannot be pulled out of the average to give

$$\overline{Q^*} = U^{m_U} L^{m_L}\, \overline{Q}. \tag{133}$$

Thus, to reduce X' so that the velocity and length scale factors can be pulled out of ensemble averages, we introduce scale relaxation, which lets the moments fluctuate in return for reduced fluctuations in the velocity and length scales.

Throughout this subsection $\mathcal{X} = \ln X$ for $X \in \{U, L\}$, as in (49), and a statement made for $\mathcal{X}$ holds for $\mathcal{U}$ and $\mathcal{L}$ alike.

The miss. The superscript con, for *constrained*, marks a quantity fixed by meeting both constraints exactly. For a field $\tilde{u}$, the *constrained ratios* $\lambda_X^{\mathrm{con}}[\tilde{u}]$ are the ratios that put $\tilde{u}$ exactly back on the targets E_0, E_2 in a single rescaling. They are the ratios the uncontrolled scaling step applies. The *miss*

$$\mu_X[\tilde{u}] = \ln \lambda_X^{\mathrm{con}}[\tilde{u}] \tag{134}$$

is therefore the amount by which $\mathcal{X}$ would have to advance to put $\tilde{u}$ back on the targets with no physics in between. By (128) the moments of $\tilde{u}$ miss their targets by

$$\ln \frac{E_0[\tilde{u}]}{E_0} = 2\mu_U + 3\mu_L, \qquad \ln \frac{E_2[\tilde{u}]}{E_2} = 2\mu_U + 5\mu_L, \tag{135}$$

so both are met if and only if $\mu_U = \mu_L = 0$.

The *constrained log-scale*

$$\mathcal{X}^{\mathrm{con}} = \mathcal{X} + \ln \lambda_X^{\mathrm{con}}[\tilde{u}] = \mathcal{X} + \mu_X \tag{136}$$

is the log-scale the frame would have after applying the constrained ratios, that is, if the field were put back on the targets at once. It is therefore unchanged by a scaling step: the step adds $\ln \lambda_X$ to $\mathcal{X}$ and removes it from μ_X. A physics step leaves $\mathcal{X}$ unchanged but alters the moments of the field, and with them μ_X. Hence $\mathcal{X}^{\mathrm{con}}$ moves only with the physics, and the scaling step only decides how much of the gap $\mathcal{X}^{\mathrm{con}} - \mathcal{X} = \mu_X$ it closes.

The relaxation. Under scale relaxation each log-scale obeys

$$\frac{\mathrm{d}\mathcal{X}}{\mathrm{d}t} = \gamma \left(\mathcal{X}^{\mathrm{tgt}} - \mathcal{X} \right), \tag{137}$$

with a target $\mathcal{X}^{\mathrm{tgt}}$ fixed below and a single gain $\gamma > 0$, a run parameter, shared by $\mathcal{U}$ and $\mathcal{L}$.

To choose the target we treat t as continuous: let $\Delta t \to 0$ with γ held fixed, and assume the physics moves $\mathcal{X}^{\mathrm{con}}$ by $O(\Delta t)$ per step, so that the rate $\sigma_X^{\mathrm{phys}} = \mathrm{d}\mathcal{X}^{\mathrm{con}}/\mathrm{d}t$ at which it does so has a limit. By (136) the miss then gains what the physics adds and loses what the scaling applies,

$$\dot{\mu}_X = \sigma_X^{\mathrm{phys}} - \dot{\mathcal{X}}. \tag{138}$$

The target. Differentiating the balance (138) once,

$$\ddot{\mu}_X = \dot{\sigma}_X^{\text{phys}} - \ddot{\mathcal{X}}.$$

The second derivative of the log-scale follows from differentiating the relaxation (137), in which γ is constant,

$$\ddot{\mathcal{X}} = \gamma\left(\dot{\mathcal{X}}^{\text{tgt}} - \dot{\mathcal{X}}\right) = \gamma\left(\dot{\mathcal{X}}^{\text{tgt}} - \sigma_X^{\text{phys}} + \dot{\mu}_X\right).$$

Substituting this into the first and moving the term $\gamma\dot{\mu}_X$ to the left-hand side,

$$\ddot{\mu}_X + \gamma\dot{\mu}_X = \dot{\sigma}_X^{\text{phys}} - \gamma\left(\dot{\mathcal{X}}^{\text{tgt}} - \sigma_X^{\text{phys}}\right) \tag{139}$$

for any target. We define the target as

$$\mathcal{X}^{\text{tgt}} = \mathcal{X}^{\text{con}} + \frac{I_X}{T_\mu}, \qquad \dot{I}_X = \mu_X. \tag{140}$$

Dynamics of the miss. With the target (140), (139) becomes

$$\ddot{\mu}_X + \gamma\dot{\mu}_X + \frac{\gamma}{T_\mu}\,\mu_X = \dot{\sigma}_X^{\text{phys}}, \tag{141}$$

and, writing $\mu_X = \mathcal{X}^{\text{con}} - \mathcal{X}$, the log-scale itself obeys

$$\ddot{\mathcal{X}} + \gamma\dot{\mathcal{X}} + \frac{\gamma}{T_\mu}\,\mathcal{X} = \gamma\,\dot{\mathcal{X}}^{\text{con}} + \frac{\gamma}{T_\mu}\,\mathcal{X}^{\text{con}}. \tag{142}$$

Both are damped oscillators. Matching them to the standard form $\ddot{y}+2\zeta\omega_0\dot{y}+\omega_0^2 y$ gives the natural frequency $\omega_0 = \sqrt{\gamma/T_\mu}$ and the damping ratio $\zeta = \gamma/(2\omega_0) = \frac{1}{2}\sqrt{\gamma T_\mu}$. Critical damping, $\zeta = 1$, requires $\gamma T_\mu = 4$, so we take

$$T_\mu = \frac{4}{\gamma}, \tag{143}$$

independently of Δt, and then $\omega_0 = \gamma/2$. Choosing T_μ on the scale of the step instead, $T_\mu = \Delta t$, would give $\zeta = \frac{1}{2}\sqrt{\gamma\Delta t} \ll 1$, an oscillation at the frequency $\sqrt{\gamma/\Delta t}$ that grows as the step is refined.

The update. The integral in (140) is accumulated from the miss of the stored field at the start of each step,

$$I_{X,k} = I_{X,k-1} + \mu_{X,k}\,\Delta t, \qquad \mu_{X,k} = \mu_X[u_k], \tag{144}$$

starting from $I_{X,-1} = T_\mu\sigma_{X,\text{init}}/\gamma$, with the initial rates $\sigma_{U,\text{init}}, \sigma_{L,\text{init}}$ run parameters. Since $\mu_{X,0} = 0$ (§2.2), the scales start out at the rates $\gamma(\mu_{X,0} + I_{X,0}/T_\mu) = \sigma_{X,\text{init}}$: zero initial rates start them from rest, and $\sigma_{L,\text{init}} - \sigma_{U,\text{init}}$ is the initial clock rate.

Within step k, ConstrainedScale receives the evolved field $\tilde{u}^{(\text{P})}$ at the log-scales $\mathcal{X}_k + \ln\lambda_X^{\text{pred}}$, where $\ln\lambda_X^{\text{pred}}$ is the log-ratio already applied by the predicted half step, $G_X(\frac{1}{2}, \beta_X; k)$ under Strang and 0 under LieTrotter (§2.7). Its constrained ratios are the misses $\mu_X^{\text{P}} = \mu_X[\tilde{u}^{(\text{P})}]$, so by (136) $\mathcal{X}_k^{\text{con}} = \mathcal{X}_k + \ln\lambda_X^{\text{pred}} + \mu_X^{\text{P}}$. Holding the target at $\mathcal{X}_k^{\text{tgt}} = \mathcal{X}_k^{\text{con}} + I_{X,k}/T_\mu$ over the step, (137) is integrated exactly from $\mathcal{X}_k$,

$$\mathcal{X}_{k+1} = \mathcal{X}_k^{\text{tgt}} + \left(\mathcal{X}_k - \mathcal{X}_k^{\text{tgt}}\right) e^{-\gamma\Delta t}, \tag{145}$$

so that the full step relaxes as prescribed whatever the predicted half step applied. The closing scaling step applies the remainder, $\ln\lambda_X = \mathcal{X}_{k+1} - \mathcal{X}_k - \ln\lambda_X^{\text{pred}}$, or

$$\ln\lambda_X = \left(1 - e^{-\gamma\Delta t}\right)\left(\ln\lambda_X^{\text{pred}} + \mu_X^{\text{P}} + \frac{I_{X,k}}{T_\mu}\right) - \ln\lambda_X^{\text{pred}}. \tag{146}$$

With $I_{X,k} = 0$ and $\gamma\Delta t \to \infty$ these reduce to the constrained ratios μ_X^{P} of the uncontrolled step. As with the projection inside Scale, the next step measures the misses afresh, so whatever the rescaled field misses is carried forward through $\mu_{X,k+1}$ and not lost.

6 Statistics and budgets in the self-similar frame

This section sets out the post-processing shared by the Navier–Stokes cases: how statistics of the self-similar fields relate to those of the physical flow (§6.1), and the budgets of $\overline{k}$ (§6.2) and $\overline{\epsilon}$ (§6.3) in their dimensional, self-similar and isotropic forms, exact and modeled. Decaying homogeneous turbulence (§7), the integrated budgets of the puff (§8), and the pointwise budgets of the puff (§9) are special cases.

6.1 Post-processing

This subsection relates statistics of the self-similar fields to those of the physical flow, defines the dissipation rate, and sets out the steps shared by all the budgets that follow.

6.1.1 Scaling of point statistics

Let Q^* be a pointwise quantity that nondimensionalizes as

$$Q^*(L\mathbf{x}, t^*) = U^{m_U} L^{m_L}\, Q(\mathbf{x}, t), \tag{147}$$

with U and L given by the power-law ansatz (31). The origin is the only point fixed in both frames ($n = N/2$ for even N); elsewhere (147) relates the two fields along the ray $\mathbf{x}^* \propto L$, not at a fixed physical position.

Let $\overline{\cdot}$ denote the average over an ensemble of initial conditions. We assume that

1. $U(t^*)$ and $L(t^*)$, i.e. A_X, α_X, β_X, are the same for every member of the ensemble;
2. in the self-similar frame the statistics are stationary and isotropic about the origin, so that $\overline{Q}$ depends only on $r = |\mathbf{x}|$;
3. the flow is ergodic in the self-similar frame.

By the first two,

$$\overline{Q^*}(r^*, t^*) = U^{m_U} L^{m_L}\, \overline{Q}(r^*/L). \tag{148}$$

By isotropy every point on the sphere $|\mathbf{x}| = r$ samples the same statistic, so within a snapshot we average over that sphere,

$$\langle Q \rangle_r(t) = \frac{1}{4\pi} \oint_{|\hat{\mathbf{n}}|=1} Q(r\hat{\mathbf{n}}, t)\, \mathrm{d}\Omega(\hat{\mathbf{n}}). \tag{149}$$

The sphere does not pass through grid points, so (149) is evaluated on the spectral interpolant of Q, where it reduces to a weighted sum over the Fourier modes (Appendix B); at $r = 0$ it is the point value $Q(0, t)$. By ergodicity the ensemble average is a time average,

$$\overline{Q}(r) = \frac{1}{T} \int_0^T \langle Q \rangle_r(t)\, \mathrm{d}t, \tag{150}$$

and a single long simulation suffices. In practice T is finite and the initial transient is discarded, so (150) only estimates $\overline{Q}(r)$; Appendix C describes its statistical error. The spherical mean reduces this error relative to a single point, the more so at larger r, where the sphere spans more correlation lengths, and not at all at $r = 0$.

6.1.2 The dissipation rate

With zero ensemble-mean velocity, the equation for the Reynolds stress $\overline{u_i^* u_j^*}$ contains the sink $-\epsilon_{ij}^*$, with the dissipation tensor [2]

$$\epsilon_{ij}^* = 2\nu^* \overline{\frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}}. \tag{151}$$

We report half its trace,

$$\epsilon^* = \tfrac{1}{2}\epsilon_{ii}^* = \nu^* \overline{\frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_i^*}{\partial x_k^*}}, \tag{152}$$

the sink in the equation for $\overline{k^*}$, $k^* = \frac{1}{2}u_i^* u_i^*$.

The LES resolves only $\tilde{u}_i^*$, so we estimate the filtered integrand. It is the resolved part $2\nu^*\,(\partial\tilde{u}_i^*/\partial x_k^*)(\partial\tilde{u}_j^*/\partial x_k^*)$ plus $2\nu^*$ times the unclosed subgrid correlation $\mathcal{G}_{\Delta^*}\big((\partial u_i^*/\partial x_k^*)(\partial u_j^*/\partial x_k^*)\big) - (\partial\tilde{u}_i^*/\partial x_k^*)(\partial\tilde{u}_j^*/\partial x_k^*)$. Consistently with the eddy-viscosity closure (88), we close the latter by gradient transport as $(\nu_{\mathrm{t}}^*/\nu^*)\,(\partial\tilde{u}_i^*/\partial x_k^*)(\partial\tilde{u}_j^*/\partial x_k^*)$, so that

$$\mathcal{G}_{\Delta^*}\left(2\nu^*\,\frac{\partial u_i^*}{\partial x_k^*}\frac{\partial u_j^*}{\partial x_k^*}\right) \approx 2\,(\nu^* + \nu_{\mathrm{t}}^*)\,\frac{\partial\tilde{u}_i^*}{\partial x_k^*}\frac{\partial\tilde{u}_j^*}{\partial x_k^*}. \tag{153}$$

The filter commutes with the ensemble average, so the average of half the trace of (153) estimates (152). With $\nu_{\mathrm{t}}^* = UL\,\nu_{\mathrm{t}}$, $\nu^* = UL/Re$, and $\partial\tilde{u}_i^*/\partial x_k^* = (U/L)\,\partial\tilde{u}_i/\partial x_k$,

$$\epsilon^* = \frac{U^3}{L}\,\epsilon, \qquad \epsilon = \left(\frac{1}{Re} + \nu_{\mathrm{t}}\right)\frac{\partial\tilde{u}_i}{\partial x_k}\frac{\partial\tilde{u}_i}{\partial x_k}, \tag{154}$$

of the form (147) with $m_U = 3$, $m_L = -1$. For $1/Re = 0$ only the subgrid part remains, and by (91) and (96)

$$\epsilon = \nu_{\mathrm{t}}\,\frac{\partial\tilde{u}_i}{\partial x_k}\frac{\partial\tilde{u}_i}{\partial x_k}, \qquad \nu_{\mathrm{t}} = \begin{cases} (C_s\Delta)^2|\tilde{S}|, & \text{Smagorinsky model,} \\ C_\nu\Delta^{4/3}, & \text{self-similar constant eddy viscosity.} \end{cases} \tag{155}$$

6.1.3 Scaling rates in the stationary state

Take $\beta_L - \beta_U = 1$ and a shared offset $\alpha_U = \alpha_L = \alpha$ in (31). The clock (83) is then $\tau = L/U = c_\tau(t^* + \alpha)$ with $c_\tau = A_L/A_U$, and by (51) the rates (4) are constant,

$$\sigma_X = \beta_X\,\frac{\tau}{t^* + \alpha} = c_\tau\beta_X, \qquad \sigma_L - \sigma_U = c_\tau. \tag{156}$$

By (84), c_τ is the rate at which $\ln\tau$ grows in t. In the budgets below we take the rates constant, with c_τ and β_X from the straight-line fit of §3.2. This is exact when the scales are prescribed by the power law; under the scale relaxation of §5.8 the instantaneous clock rate $\sigma_{L,k} - \sigma_{U,k}$ (52) fluctuates about c_τ, and the relaxation keeps it slowly varying.

6.1.4 Budgets in the self-similar frame

Notation and assumptions. From here on every field is a resolved field of the LES, and we drop the tilde of §5.1: u_i and p denote $\tilde{u}_i$ and $\tilde{p}$, and $k = \frac{1}{2}u_i u_i$. The budgets are formed under the following assumptions:

1. $1/Re = 0$, and the subgrid model is the self-similar constant eddy viscosity (96), so that $\nu_{\mathrm{t}}^* = UL\,\nu_{\mathrm{t}}$ is uniform in space but not in time; the budgets are not formed under the Smagorinsky model;
2. the ensemble-mean velocity vanishes, $\overline{u_i} = 0$, so u_i is the fluctuating velocity, and all production by and advection with the mean flow vanish;
3. we work inside the sphere $r \le r_0$, where $W = 0$, so the sponge does not enter.

The subgrid stress term is then the Laplacian (98), the filtered momentum equation (76) becomes

$$\frac{\partial u_i^*}{\partial t^*} + u_k^*\frac{\partial u_i^*}{\partial x_k^*} = -\frac{1}{\rho}\frac{\partial p^*}{\partial x_i^*} + \nu_{\mathrm{t}}^*\,\frac{\partial^2 u_i^*}{\partial x_k^*\partial x_k^*}, \tag{157}$$

and by (153) and (154) the dissipation rate is

$$\epsilon^* = \nu_{\mathrm{t}}^*\,\frac{\partial u_i^*}{\partial x_k^*}\frac{\partial u_i^*}{\partial x_k^*}, \qquad \epsilon = \nu_{\mathrm{t}}\,\frac{\partial u_i}{\partial x_k}\frac{\partial u_i}{\partial x_k}, \qquad \epsilon^* = \frac{U^3}{L}\,\epsilon. \tag{158}$$

Transformation to the self-similar frame. Each budget has the form $\partial\overline{Q^*}/\partial t^* = \sum_n S_n^*$ for a mean statistic scaling as (147). With the substitutions $u_i^* = U u_i$, $x_j^* = L x_j$, $p^* = \rho U^2 p$, and $\nu_{\mathrm{t}}^* = U L\,\nu_{\mathrm{t}}$, each source term becomes $S_n^* = (U^{m_U} L^{m_L}/\tau)\, S_n$, where S_n is the same expression without stars and with $\rho = 1$, and $\tau = L/U$ is the clock (83). The chain rule (6), with the amplitude $U^{m_U} L^{m_L}$ (rate $m_U\sigma_U + m_L\sigma_L$) in place of U, gives the same factor on the left, so

$$\frac{\partial \overline{Q}}{\partial t} = \sigma_L\, x_j \frac{\partial \overline{Q}}{\partial x_j} - (m_U\sigma_U + m_L\sigma_L)\,\overline{Q} + \sum_n S_n. \tag{159}$$

The first term advects $\overline{Q}$ radially as the frame dilates; the second is the rescaling of its amplitude. In the stationary state the left-hand side vanishes and the rates are the constants (156).

Radial form. Under isotropy a mean scalar f depends only on r, and a mean vector is radial, $F_j = n_j F(r)$ with $n_j = x_j/r$. Then

$$x_j \frac{\partial f}{\partial x_j} = r\frac{\mathrm{d}f}{\mathrm{d}r}, \qquad \frac{\partial F_j}{\partial x_j} = \frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 F\right), \qquad \frac{\partial^2 f}{\partial x_j \partial x_j} = \nabla_r^2 f \equiv \frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2 \frac{\mathrm{d}f}{\mathrm{d}r}\right). \tag{160}$$

At $r = 0$ each of these has the form $0/0$, and its value follows from regularity. A smooth function of r alone is even, $f = f(0) + \frac{1}{2}f''(0)\, r^2 + O(r^4)$, and smoothness of $x_j F/r$ requires F to be odd, $F = F'(0)\, r + O(r^3)$, so that

$$x_j \frac{\partial f}{\partial x_j}\bigg|_0 = 0, \qquad \frac{\partial F_j}{\partial x_j}\bigg|_0 = 3F'(0), \qquad \nabla_r^2 f\big|_0 = 3f''(0), \tag{161}$$

and $f'(0) = 0$. A flux thus vanishes at the origin, where there is no preferred direction, but its divergence does not. When the profiles are evaluated numerically, (161) replaces (160) at $r = 0$. Every term of the resulting radial budgets is estimated independently, so their residuals measure the statistical and discretization error of the profiles.

6.2 Kinetic energy budget

6.2.1 Exact budget

The dimensional kinetic energy budget is

$$\frac{\partial \overline{k^*}}{\partial t^*} = T_k^* + \Pi_k^* + D_k^* - \overline{\epsilon^*}, \tag{162}$$

with

$$\begin{aligned} T_k^* &= -\frac{\partial}{\partial x_j^*}\,\overline{u_j^* k^*} && \text{turbulent transport,} \\ \Pi_k^* &= -\frac{1}{\rho}\frac{\partial}{\partial x_j^*}\,\overline{u_j^* p^*} && \text{pressure transport,} \\ D_k^* &= \nu_{\mathrm{t}}^* \frac{\partial^2 \overline{k^*}}{\partial x_j^* \partial x_j^*} && \text{viscous diffusion,} \end{aligned} \tag{163}$$

and $\overline{\epsilon^*}$ (158).

Self-similar frame. The kinetic energy and the dissipation scale as

$$\overline{k^*} = U^2\,\overline{k}, \qquad \overline{\epsilon^*} = \frac{U^3}{L}\,\overline{\epsilon}, \tag{164}$$

by (122) and (158), so with $m_U = 2$, $m_L = 0$, (159) gives

$$\frac{\partial \overline{k}}{\partial t} = \sigma_L\, x_j \frac{\partial \overline{k}}{\partial x_j} - 2\sigma_U\,\overline{k} + T_k + \Pi_k + D_k - \overline{\epsilon}, \tag{165}$$

with T_k, Π_k, D_k of the form (163) without stars and with $\rho = 1$.

Isotropic form. The fluxes $\overline{u_j k}$ and $\overline{u_j p}$ are radial, so by (160)

$$T_k = -\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\,\overline{u_r k}\right), \qquad \Pi_k = -\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\,\overline{u_r p}\right), \qquad D_k = \nu_\mathrm{t}\,\nabla_r^2\overline{k}, \tag{166}$$

and in the stationary state (165) becomes

$$0 = \sigma_L\, r\frac{\mathrm{d}\overline{k}}{\mathrm{d}r} - 2\sigma_U\,\overline{k} + T_k + \Pi_k + D_k - \overline{\epsilon}. \tag{167}$$

It is the budget used by the k–ϵ model of §6.3.2.

6.2.2 Modeling terms

In (162) the dissipation $\overline{\epsilon^*}$ is the variable of the ϵ equation, and the viscous diffusion D_k^* is closed and kept exactly, with the self-similar eddy viscosity ν_t^* uniform in space and hence outside the derivatives. Only the transport $T_k^* + \Pi_k^*$ is unclosed. Instead of the gradient diffusion $\partial_j^*\left((\nu_\epsilon^*/\sigma_k)\,\partial_j^*\overline{k^*}\right)$ of the standard model [2], we close it with the same radial drift as the transport of $\overline{\epsilon^*}$ (§6.3.2), so that both equations of the model share one transport closure:

$$T_k^* + \Pi_k^* \approx -C_{k\mathrm{T}}\frac{\partial}{\partial x_j^*}\left(v_j^*\,\overline{k^*}\right), \qquad v_j^* = \frac{1}{L}\frac{\mathrm{d}L}{\mathrm{d}t^*}\,x_j^*, \tag{168}$$

with x_j^* measured from the centre of the domain and $C_{k\mathrm{T}}$ a constant to be fitted. As for $\overline{\epsilon^*}$, $C_{k\mathrm{T}} = 1$ carries $\overline{k^*}$ along with the expanding profiles, the drift vanishes in homogeneous turbulence, and the model describes the self-similar state rather than predicting its spreading rate. Unlike ϵ^*, k^* does not contain ν_t^*, so there is no eddy-viscosity change to model and $C_{k\mathrm{T}}$ needs no counterpart of $C_{\epsilon\mathrm{T}\nu}$. The modeled dimensional budget is

$$\frac{\partial\overline{k^*}}{\partial t^*} = \nu_\mathrm{t}^*\frac{\partial^2\overline{k^*}}{\partial x_j^*\partial x_j^*} - C_{k\mathrm{T}}\frac{\partial}{\partial x_j^*}\left(v_j^*\,\overline{k^*}\right) - \overline{\epsilon^*}. \tag{169}$$

Self-similar frame. By (164) and $v_j^* = U\,\sigma_L x_j$ (184), the drift term carries $U\cdot U^2/L = U^3/L$, the factor of every term of (162), so (159) gives

$$\frac{\partial\overline{k}}{\partial t} = \sigma_L\,x_j\frac{\partial\overline{k}}{\partial x_j} - 2\sigma_U\,\overline{k} + \nu_\mathrm{t}\frac{\partial^2\overline{k}}{\partial x_j\partial x_j} - C_{k\mathrm{T}}\,\sigma_L\frac{\partial}{\partial x_j}\left(x_j\,\overline{k}\right) - \overline{\epsilon}. \tag{170}$$

Together with (185) it forms the closed k–ϵ model of the puff.

Isotropic form. With $\partial x_j/\partial x_j = 3$, the divergence of the drift flux is

$$C_{k\mathrm{T}}\,\sigma_L\frac{\partial}{\partial x_j}\left(x_j\,\overline{k}\right) = C_{k\mathrm{T}}\,\sigma_L\left(3\overline{k} + x_j\frac{\partial\overline{k}}{\partial x_j}\right),$$

so in the stationary state (170) becomes

$$0 = (1 - C_{k\mathrm{T}})\,\sigma_L\,x_j\frac{\partial\overline{k}}{\partial x_j} - (2\sigma_U + 3C_{k\mathrm{T}}\,\sigma_L)\,\overline{k} + \nu_\mathrm{t}\frac{\partial^2\overline{k}}{\partial x_j\partial x_j} - \overline{\epsilon}, \tag{171}$$

in which, under isotropy, every term is a function of r alone and is evaluated as a spherical mean (149). As for $\overline{\epsilon}$, the drift cancels a fraction $C_{k\mathrm{T}}$ of the advection by the dilating frame and adds the dilution $3C_{k\mathrm{T}}\,\sigma_L\,\overline{k}$. Comparing the modeled transport $-C_{k\mathrm{T}}\,\sigma_L\left(3\overline{k} + x_j\,\partial\overline{k}/\partial x_j\right)$ with the measured $T_k + \Pi_k$ tests the closure directly. At the origin $x_j\,\partial\overline{k}/\partial x_j = 0$ (161), so the modeled transport reduces to $-3C_{k\mathrm{T}}\,\sigma_L\,\overline{k}(0)$, which is negative for $\sigma_L > 0$: the drift drains energy from the centre.

6.3 ϵ budget

In the stationary state the budget of $\overline{\epsilon}$ reduces to a balance (177) between independently measured profiles, which tests them, and its dimensional form (172) is the starting point of the model of §6.3.2.

6.3.1 Exact budget

Dimensional budget. Since ν_{t}^* is uniform in space but not in time, differentiating (158) gives

$$\frac{\partial \overline{\epsilon^*}}{\partial t^*} = \frac{1}{\nu_{\mathrm{t}}^*}\frac{\mathrm{d}\nu_{\mathrm{t}}^*}{\mathrm{d}t^*}\,\overline{\epsilon^*} + \nu_{\mathrm{t}}^*\,\frac{\partial}{\partial t^*}\,\overline{\frac{\partial u_i^*}{\partial x_k^*}\frac{\partial u_i^*}{\partial x_k^*}}.$$

In the second term the viscosity enters only as the instantaneous coefficient of the viscous term of (157), so the derivation of [3] applies unchanged, with ν^* replaced by ν_{t}^*. For zero mean velocity,

$$\frac{\partial \overline{\epsilon^*}}{\partial t^*} = \frac{1}{\nu_{\mathrm{t}}^*}\frac{\mathrm{d}\nu_{\mathrm{t}}^*}{\mathrm{d}t^*}\,\overline{\epsilon^*} + P_\epsilon^* + T_\epsilon^* + \Pi_\epsilon^* + D_\epsilon^* - \Upsilon^*, \tag{172}$$

with

$$\begin{aligned}
P_\epsilon^* &= -2\nu_{\mathrm{t}}^*\,\overline{\frac{\partial u_i^*}{\partial x_k^*}\frac{\partial u_i^*}{\partial x_m^*}\frac{\partial u_k^*}{\partial x_m^*}} && \text{turbulent production,}\\
T_\epsilon^* &= -\frac{\partial}{\partial x_j^*}\,\overline{u_j^*\epsilon^*} && \text{turbulent transport,}\\
\Pi_\epsilon^* &= -\frac{2\nu_{\mathrm{t}}^*}{\rho}\,\frac{\partial}{\partial x_j^*}\,\overline{\frac{\partial p^*}{\partial x_m^*}\frac{\partial u_j^*}{\partial x_m^*}} && \text{pressure transport,}\\
D_\epsilon^* &= \nu_{\mathrm{t}}^*\frac{\partial^2\overline{\epsilon^*}}{\partial x_j^*\partial x_j^*} && \text{viscous diffusion,}\\
\Upsilon^* &= 2\,(\nu_{\mathrm{t}}^*)^2\,\overline{\frac{\partial^2 u_i^*}{\partial x_k^*\partial x_m^*}\frac{\partial^2 u_i^*}{\partial x_k^*\partial x_m^*}} && \text{dissipation of } \epsilon^*.
\end{aligned} \tag{173}$$

P_ϵ^* is the production P_ϵ^4 of [3]; their mean-advection term and the productions P_ϵ^1–P_ϵ^3 involve the mean velocity and vanish. The eddy-viscosity change is the only term the LES adds.

Self-similar frame. Here $m_U = 3$, $m_L = -1$, and each of the five terms (173) carries the factor $(U^3/L)/\tau = U^4/L^2$, e.g. $P_\epsilon^* = (UL)(U/L)^3\,P_\epsilon$. The eddy-viscosity change is not of this form: with $\nu_{\mathrm{t}}^* \propto UL$ and $\mathrm{d}t^*/\mathrm{d}t = \tau$,

$$\frac{1}{\nu_{\mathrm{t}}^*}\frac{\mathrm{d}\nu_{\mathrm{t}}^*}{\mathrm{d}t^*} = \frac{\sigma_U + \sigma_L}{\tau}, \tag{174}$$

so it contributes $(U^4/L^2)(\sigma_U+\sigma_L)\,\overline{\epsilon}$. By (159),

$$\frac{\partial \overline{\epsilon}}{\partial t} = \sigma_L\,x_j\frac{\partial \overline{\epsilon}}{\partial x_j} - (3\sigma_U - \sigma_L)\,\overline{\epsilon} + (\sigma_U+\sigma_L)\,\overline{\epsilon} + P_\epsilon + T_\epsilon + \Pi_\epsilon + D_\epsilon - \Upsilon, \tag{175}$$

with

$$\begin{aligned}
&P_\epsilon = -2\nu_{\mathrm{t}}\,\overline{\frac{\partial u_i}{\partial x_k}\frac{\partial u_i}{\partial x_m}\frac{\partial u_k}{\partial x_m}}, \quad T_\epsilon = -\frac{\partial}{\partial x_j}\,\overline{u_j\epsilon}, && \Pi_\epsilon = -2\nu_{\mathrm{t}}\,\frac{\partial}{\partial x_j}\,\overline{\frac{\partial p}{\partial x_m}\frac{\partial u_j}{\partial x_m}},\\
&D_\epsilon = \nu_{\mathrm{t}}\frac{\partial^2\overline{\epsilon}}{\partial x_j\partial x_j}, && \Upsilon = 2\nu_{\mathrm{t}}^2\,\overline{\frac{\partial^2 u_i}{\partial x_k\partial x_m}\frac{\partial^2 u_i}{\partial x_k\partial x_m}}.
\end{aligned} \tag{176}$$

The first two terms on the right of (175) arise from the time dependence of the scales, as for any statistic; the third arises from the time dependence of the viscosity and vanishes for a viscosity constant in physical space. We keep them separate, since the model of §6.3.2 treats them differently; together they are $-2(\sigma_U-\sigma_L)\,\overline{\epsilon} = 2c_\tau\,\overline{\epsilon}$ by (156). In the stationary state the profiles satisfy

$$0 = \sigma_L\,r\frac{\mathrm{d}\overline{\epsilon}}{\mathrm{d}r} - (3\sigma_U - \sigma_L)\,\overline{\epsilon} + (\sigma_U+\sigma_L)\,\overline{\epsilon} + P_\epsilon + T_\epsilon + \Pi_\epsilon + D_\epsilon - \Upsilon, \tag{177}$$

each term a function of r alone.

6.3.2 Modeling terms

The production, dissipation, and transport terms of (172) are unclosed. We close the production and dissipation in terms of $\overline{k^*}$ and $\overline{\epsilon^*}$, as in the standard k–ϵ model [2]; the transport as a radial drift with the self-similar expansion rather than as gradient diffusion; and the eddy-viscosity change with the same two closure forms. The viscous diffusion is closed and kept exactly.

Production and dissipation. With zero mean velocity, P_ϵ^* and Υ^* are the two large terms of the budget and nearly balance. We model their difference as

$$P_\epsilon^* - \Upsilon^* \approx -C_{\epsilon 2}\,\frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}}. \tag{178}$$

Transport. The standard model closes the transport as gradient diffusion, $\partial_j^*\left((\nu_\epsilon^*/\sigma_\epsilon)\,\partial_j^*\overline{\epsilon^*}\right)$ with $\nu_\epsilon^* = C_\mu(\overline{k^*})^2/\overline{\epsilon^*}$. In the puff this fails. The radial flux of $\bar\epsilon$, obtained by integrating the measured $T_\epsilon + \Pi_\epsilon$ outward from the origin, already points outward at the origin, where $\bar\epsilon$ has a shallow local minimum, so near the centre the transport runs up the gradient; and the flux stays large on the outer shoulder, where $\bar\epsilon$ is nearly uniform. No positive diffusivity reproduces either. The flux is instead close to that of a radial drift with the self-similar expansion of the flow, so we model

$$T_\epsilon^* + \Pi_\epsilon^* \approx -C_{\epsilon\mathrm{T}}\,\frac{\partial}{\partial x_j^*}\left(v_j^*\,\overline{\epsilon^*}\right), \qquad v_j^* = \frac{1}{L}\frac{\mathrm{d}L}{\mathrm{d}t^*}\,x_j^*, \tag{179}$$

with x_j^* measured from the centre of the domain, about which the frame dilates. Since v_j^* is the velocity of a point fixed in the self-similar frame, $C_{\epsilon\mathrm{T}} = 1$ carries $\overline{\epsilon^*}$ along with the expanding profiles, and $C_{\epsilon\mathrm{T}} < 1$ (> 1) lets it lag behind (run ahead). The model is not local: it uses the position of the centre and the global rate $(1/L)\,\mathrm{d}L/\mathrm{d}t^*$. It thus describes the self-similar state rather than predicting its spreading rate, which suffices here: the rates are measured, and the purpose of the model is to account for the transport so that (178) can be tested on its own. In homogeneous turbulence, $\sigma_L = 0$, the drift vanishes, as does the exact transport. The viscous diffusion keeps its exact form (173), with the self-similar eddy viscosity ν_t^* uniform in space and hence outside the derivatives, so (172) becomes

$$\frac{\partial\overline{\epsilon^*}}{\partial t^*} = \frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*}\,\overline{\epsilon^*} + \nu_\mathrm{t}^*\frac{\partial^2\overline{\epsilon^*}}{\partial x_j^*\partial x_j^*} - C_{\epsilon\mathrm{T}}\,\frac{\partial}{\partial x_j^*}\left(v_j^*\,\overline{\epsilon^*}\right) - C_{\epsilon 2}\,\frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}}. \tag{180}$$

Eddy-viscosity change. The first term on the right of (180) is closed, but it is an artifact of the eddy viscosity scaling with UL: a flow with a viscosity constant in physical space has no such term. Solving the stationary kinetic energy budget (165) for σ_U at each point gives an identity for the term $(\sigma_U + \sigma_L)\,\bar\epsilon$ of (175),

$$(\sigma_U + \sigma_L)\,\bar\epsilon = -\frac{1}{2}\,\frac{\bar\epsilon^{\,2}}{\overline{k}} + \frac{\bar\epsilon}{2\overline{k}}\left(\sigma_L\,x_j\frac{\partial\overline{k}}{\partial x_j} + T_k + \Pi_k + D_k\right) + \sigma_L\,\bar\epsilon. \tag{181}$$

In homogeneous turbulence only the first term survives, and the eddy-viscosity change is exactly $-\frac{1}{2}$ times the closure term. The others arise from the dilation of the frame and the transport of $\overline{k}$ and vanish for $\sigma_L = 0$; if the transport of $\overline{k}$ is, like that of $\bar\epsilon$, a drift with the expansion, and $\bar\epsilon/\overline{k}$ is nearly uniform across the puff, they are combinations of $\bar\epsilon$ and $x_j\,\partial_j\bar\epsilon$, as is the drift. We therefore model the eddy-viscosity change with both closure forms,

$$\frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*}\,\overline{\epsilon^*} \approx -C_{\epsilon\nu}\,\frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}} - C_{\epsilon\mathrm{T}\nu}\,\frac{\partial}{\partial x_j^*}\left(v_j^*\,\overline{\epsilon^*}\right), \tag{182}$$

each with its own fitted constant. For homogeneous turbulence the model is exact with $C_{\epsilon\nu} = \frac{1}{2}$; in the puff, $C_{\epsilon\nu}$ close to $\frac{1}{2}$ indicates that the drift carries the remaining terms of (181). Substituting into (180),

$$\frac{\partial\overline{\epsilon^*}}{\partial t^*} = \nu_\mathrm{t}^*\frac{\partial^2\overline{\epsilon^*}}{\partial x_j^*\partial x_j^*} - C_{\epsilon\mathrm{T,c}}\,\frac{\partial}{\partial x_j^*}\left(v_j^*\,\overline{\epsilon^*}\right) - C_{\epsilon 2,\mathrm{c}}\,\frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}}, \qquad C_{\epsilon 2,\mathrm{c}} = C_{\epsilon 2} + C_{\epsilon\nu}, \quad C_{\epsilon\mathrm{T,c}} = C_{\epsilon\mathrm{T}} + C_{\epsilon\mathrm{T}\nu}, \tag{183}$$

the k–ϵ equation for zero mean velocity with the drift transport and the constants of the constant-viscosity flow; the eddy viscosity now enters only through the diffusion. For homogeneous turbulence, $C_{\epsilon 2,\mathrm{c}} = C_{\epsilon 2} + \frac{1}{2}$. Since our ϵ^* is the sink of $\overline{k^*}$, as in [2], it is $C_{\epsilon 2,\mathrm{c}}$, not $C_{\epsilon 2}$, that compares with the standard constant.

Self-similar frame. The modeled terms scale like the exact terms they replace. By (164), $(1/L)\,\mathrm{d}L/\mathrm{d}t^* = \sigma_L/\tau$, and $\tau = L/U$,

$$\frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}} = \frac{U^4}{L^2}\,\frac{\overline{\epsilon}^{\,2}}{\overline{k}}, \qquad v_j^* = U\,v_j, \quad v_j = \sigma_L\,x_j, \tag{184}$$

so every term on the right of (183) carries U^4/L^2, and (159) gives, now without the eddy-viscosity term $(\sigma_U + \sigma_L)\,\overline{\epsilon}$,

$$\frac{\partial \overline{\epsilon}}{\partial t} = \sigma_L\,x_j\frac{\partial \overline{\epsilon}}{\partial x_j} - (3\sigma_U - \sigma_L)\,\overline{\epsilon} + \nu_\mathrm{t}\frac{\partial^2 \overline{\epsilon}}{\partial x_j \partial x_j} - C_{\epsilon\mathrm{T},\mathrm{c}}\,\sigma_L\frac{\partial}{\partial x_j}\,(x_j\,\overline{\epsilon}) - C_{\epsilon 2,\mathrm{c}}\,\frac{\overline{\epsilon}^{\,2}}{\overline{k}}, \tag{185}$$

with $\nu_\mathrm{t} = C_\nu \Delta^{4/3}$.

Isotropic form. The drift flux is radial, $F = C_{\epsilon\mathrm{T},\mathrm{c}}\,\sigma_L\,r\,\overline{\epsilon}$, so by (160) its divergence is $C_{\epsilon\mathrm{T},\mathrm{c}}\,\sigma_L\,r^{-2}\,\mathrm{d}(r^3\overline{\epsilon})/\mathrm{d}r = C_{\epsilon\mathrm{T},\mathrm{c}}\,\sigma_L(3\overline{\epsilon} + r\,\mathrm{d}\overline{\epsilon}/\mathrm{d}r)$. In the stationary state (185) therefore reduces to the ordinary differential equation

$$0 = (1 - C_{\epsilon\mathrm{T},\mathrm{c}})\,\sigma_L\,r\frac{\mathrm{d}\overline{\epsilon}}{\mathrm{d}r} - (3\sigma_U - \sigma_L + 3C_{\epsilon\mathrm{T},\mathrm{c}}\,\sigma_L)\,\overline{\epsilon} + \nu_\mathrm{t}\,\nabla_r^2\overline{\epsilon} - C_{\epsilon 2,\mathrm{c}}\,\frac{\overline{\epsilon}^{\,2}}{\overline{k}}, \tag{186}$$

with $\overline{\epsilon}'(0) = 0$. The drift cancels a fraction $C_{\epsilon\mathrm{T},\mathrm{c}}$ of the advection by the dilating frame, leaving $\overline{\epsilon}$ unadvected for $C_{\epsilon\mathrm{T},\mathrm{c}} = 1$, and adds the uniform-rate sink $3C_{\epsilon\mathrm{T},\mathrm{c}}\,\sigma_L\,\overline{\epsilon}$, the dilution of $\overline{\epsilon}$ by the expanding drift.

7 Decaying homogeneous turbulence: a post-processing thought experiment

This section applies the framework to decaying homogeneous turbulence and asks what the constant $C_{\epsilon 2}$ of the k–ϵ model computed from the self-similar simulation means for the physical flow.

Notation. The notation and assumptions are those of §6.1.4: every field is a resolved field of the LES, an overline denotes the ensemble average, and the mean kinetic energy and dissipation are $\overline{k^*}$ and $\overline{\epsilon^*}$ in physical variables and $\overline{k}$ and $\overline{\epsilon}$ in the self-similar frame, related by (164). Since the turbulence is homogeneous, these averages are uniform in space and depend on time alone, and every term of the budgets of §6.2 and §6.3 that is a divergence vanishes.

The simulated and the true flow. Statistics of two flows are compared. The *simulated* flow is the one we compute, with the self-similar constant eddy viscosity (96): it is constant in the self-similar frame, $\nu_\mathrm{t} = C_\nu\Delta^{4/3}$, and therefore time dependent in physical space, $\nu_\mathrm{t}^* = UL\,\nu_\mathrm{t}$. The *true* flow is the one whose viscosity is constant in physical space and equal to zero,

$$\nu_\mathrm{c}^* = 0, \tag{187}$$

the infinite-Reynolds-number flow that the simulation, with $1/Re = 0$, approximates. Its spectrum extends to $\kappa^* \to \infty$, and its dissipation is the rate at which energy cascades to infinitely small scales. Quantities of the true flow carry the subscript c, for constant viscosity: $\overline{k_\mathrm{c}^*}$, $\overline{\epsilon_\mathrm{c}^*}$, $C_{\epsilon 2,\mathrm{c}}$. Quantities without it belong to the simulated flow. In §7.2 we show that the $\overline{k^*}$ and $\overline{\epsilon^*}$ of the two flows differ by the same constant factor, close to one; in §7.3 we use this to relate the $C_{\epsilon 2}$ computed from the simulation to the $C_{\epsilon 2,\mathrm{c}}$ of the true flow.

7.1 Governing equation and the scaling rate

For decaying homogeneous turbulence we hold the length scale L constant, so $\sigma_L = 0$ and the scaling term $\sigma_L x_j\, \partial u_i/\partial x_j$ of (86) drops out. With zero molecular viscosity, $1/Re = 0$, and the subgrid term (98), the momentum equation is

$$\frac{\partial u_i}{\partial t} = -\sigma_U\, u_i - u_j \frac{\partial u_i}{\partial x_j} - \frac{\partial p}{\partial x_i} + C_\nu \Delta^{4/3} \frac{\partial^2 u_i}{\partial x_j \partial x_j}. \tag{188}$$

The only coefficient that can depend on time is σ_U. With the clock $\tau = L/U$ (83) and $\mathrm{d}t^*/\mathrm{d}t = \tau$,

$$\sigma_U = \frac{1}{U}\frac{\mathrm{d}U}{\mathrm{d}t} = \frac{L}{U^2}\frac{\mathrm{d}U}{\mathrm{d}t^*}. \tag{189}$$

Under the power-law ansatz $U = A_U (t^* + \alpha_U)^{\beta_U}$ (§1.5),

$$\sigma_U = \frac{\beta_U}{t^* + \alpha_U}\, \frac{L}{A_U (t^* + \alpha_U)^{\beta_U}} = \frac{\beta_U L}{A_U}\, \frac{1}{(t^* + \alpha_U)^{\beta_U + 1}}. \tag{190}$$

This is constant only for $\beta_U = -1$, the value that $\beta_L - \beta_U = 1$ (§6.1.3) requires with $\beta_L = 0$; for any other decay exponent σ_U drifts in time and the self-similar state is not stationary.

To write σ_U as a function of t, we integrate the clock,

$$\frac{\mathrm{d}t^*}{\mathrm{d}t} = \frac{L}{U} = \frac{L}{A_U}(t^* + \alpha_U)^{-\beta_U}, \qquad (t^* + \alpha_U)^{\beta_U}\, \mathrm{d}t^* = \frac{L}{A_U}\, \mathrm{d}t. \tag{191}$$

Integrating from $t^* = 0$ at $t = 0$ gives, for $\beta_U \neq -1$,

$$(t^* + \alpha_U)^{\beta_U + 1} = \alpha_U^{\beta_U + 1} + (\beta_U + 1)\frac{L}{A_U}\, t. \tag{192}$$

The left-hand side is the combination appearing in (190), so

$$\sigma_U(t) = \frac{\beta_U L/A_U}{\alpha_U^{\beta_U+1} + (\beta_U + 1)\,(L/A_U)\, t} = \frac{\sigma_{U,0}}{1 + \dfrac{\beta_U + 1}{\beta_U}\, \sigma_{U,0}\, t}, \qquad \sigma_{U,0} = \sigma_U(0) = \frac{\beta_U L}{A_U\, \alpha_U^{\beta_U+1}}, \tag{193}$$

or equivalently

$$\frac{1}{\sigma_U(t)} = \frac{1}{\sigma_{U,0}} + \frac{\beta_U + 1}{\beta_U}\, t. \tag{194}$$

7.2 Time-dependent viscosity

We now compare the simulated flow, with $\nu_\mathrm{t}^* = UL\,\nu_\mathrm{t}$, with the inviscid true flow (187). In homogeneous turbulence the kinetic energy budget (162) reduces, for the simulated and the true flow, to

$$\frac{\mathrm{d}\overline{k^*}}{\mathrm{d}t^*} = -\overline{\epsilon^*}, \qquad \frac{\mathrm{d}\overline{k_\mathrm{c}^*}}{\mathrm{d}t^*} = -\overline{\epsilon_\mathrm{c}^*}, \tag{195}$$

with $\overline{\epsilon^*}$ built from ν_t^* (158). For the true flow the product of $\nu_\mathrm{c}^* = 0$ with the diverging mean squared velocity gradient is indeterminate, so the second equation of (195) is taken as the definition of $\overline{\epsilon_\mathrm{c}^*}$.

The eddy viscosity ends the spectrum $E^*(\kappa^*)$ of the simulated flow at the Kolmogorov scale η^* (92), which (93) pins to the filter width,

$$\eta^* = C_\eta \Delta L, \tag{196}$$

fixed in time since L is constant. The true flow has no dissipative scale, $\eta_\mathrm{c}^* = 0$, and its spectrum continues to $\kappa^* \to \infty$. Modeling the two spectra as a common self-similar spectrum, cut off at $2\pi/\eta^*$ in the simulated flow and not at all in the true flow, the kinetic energies are

$$\overline{k^*} \approx \int_0^{2\pi/\eta^*} E^*(\kappa^*)\, \mathrm{d}\kappa^*, \qquad \overline{k_\mathrm{c}^*} = \overline{k^*} + k_a^*, \qquad k_a^* \approx \int_{2\pi/\eta^*}^{\infty} E^*(\kappa^*)\, \mathrm{d}\kappa^* \geq 0, \tag{197}$$

where k_a^* is the energy of the tail that the simulation is missing.

To evaluate k_a^*, assume that the spectrum is self-similar, $E^*(\kappa^*, t^*) = U^2 L\, E(\kappa^* L)$, with the power law

$$E(\kappa) = A_E(\kappa + \alpha_E)^{\beta_E}, \tag{198}$$

so that in dimensional form

$$E^*(\kappa^*) = A_E^*(\kappa^* + \alpha_E^*)^{\beta_E}, \qquad A_E^* = A_E\, U^2 L^{1+\beta_E}, \qquad \alpha_E^* = \frac{\alpha_E}{L}. \tag{199}$$

The tail integral converges for an inertial-range spectrum, $\beta_E < -1$, and with (196)

$$k_a^* = -\frac{A_E^*}{\beta_E + 1}\left(\frac{2\pi}{C_\eta \Delta L} + \alpha_E^*\right)^{\beta_E+1} = U^2 k_a, \qquad k_a = -\frac{A_E}{\beta_E + 1}\left(\frac{2\pi}{C_\eta \Delta} + \alpha_E\right)^{\beta_E+1}. \tag{200}$$

The self-similar adjustment k_a depends only on the constants of the spectrum and the grid, so it is fixed in time, as is $\overline{k}$ under the kinetic energy constraint. The adjustment is therefore a constant fraction of the simulated energy,

$$r = \frac{k_a^*}{\overline{k^*}} = \frac{k_a}{\overline{k}} = O\Big((C_\eta \Delta)^{-(\beta_E+1)}\Big) \ll 1, \tag{201}$$

small since the dissipative scale is far below L. With (195), the two flows are related by

$$\overline{k_c^*} = (1 + r)\,\overline{k^*} = (1 + r)\, U^2\, \overline{k}, \qquad \overline{\epsilon_c^*} = (1 + r)\,\overline{\epsilon^*} = (1 + r)\,\frac{U^3}{L}\,\overline{\epsilon}, \tag{202}$$

exactly within the spectral model. Two consequences follow. First, since $r \ll 1$, it suffices to post-process the simulated flow: the self-similar $\overline{k}$ and $\overline{\epsilon}$, rescaled by (164), give the kinetic energy and dissipation of the true flow to within the factor $1 + r$. Second, $\overline{k_c^*} \propto U^2$, so the true flow decays with the same exponent β_U as the simulated one, and the ratio $\overline{\epsilon_c^*}/\overline{k_c^*} = \overline{\epsilon^*}/\overline{k^*}$ is the same in both. Since $k_a^* \neq 0$ at $t^* = 0$, the two flows do not start from the same field: the initial field of the true flow is that of the simulated flow with its spectrum continued to $\kappa^* \to \infty$.

7.3 Computing $C_{\epsilon 2}$

The simulated constant. In homogeneous turbulence with $\sigma_L = 0$, the kinetic energy budget (165) becomes

$$\frac{\mathrm{d}\overline{k}}{\mathrm{d}t} = -2\sigma_U\,\overline{k} - \overline{\epsilon}, \qquad \overline{\epsilon} = C_\nu \Delta^{4/3}\,\overline{\frac{\partial u_i}{\partial x_m}\frac{\partial u_i}{\partial x_m}}, \tag{203}$$

and the dissipation budget (175) becomes

$$\frac{\mathrm{d}\overline{\epsilon}}{\mathrm{d}t} = -3\sigma_U\,\overline{\epsilon} + \sigma_U\,\overline{\epsilon} + P_\epsilon - \Upsilon, \tag{204}$$

with P_ϵ and Υ given by (176). As in (175) with $\sigma_L = 0$, the two terms in σ_U have different origins. The first, $-3\sigma_U\,\overline{\epsilon}$, arises from the time-dependent velocity scale: it is the rescaling of $\overline{\epsilon^*} = (U^3/L)\,\overline{\epsilon}$. The second, $+\sigma_U\,\overline{\epsilon}$, arises from the time-dependent viscosity: it is the eddy-viscosity change (174), which is absent at constant viscosity. The constant $C_{\epsilon 2}$ is defined, as in (178), by closing $P_\epsilon - \Upsilon$ alone, so with (184)

$$\frac{\mathrm{d}\overline{\epsilon}}{\mathrm{d}t} = -3\sigma_U\,\overline{\epsilon} + \sigma_U\,\overline{\epsilon} - C_{\epsilon 2}\,\frac{\overline{\epsilon}^{\,2}}{\overline{k}}. \tag{205}$$

The kinetic energy constraint holds $\overline{k}$ fixed, $\mathrm{d}\overline{k}/\mathrm{d}t = 0$, so (203) gives

$$\overline{\epsilon} = -2\sigma_U\,\overline{k}, \qquad \frac{\mathrm{d}\overline{\epsilon}}{\mathrm{d}t} = -2\overline{k}\,\frac{\mathrm{d}\sigma_U}{\mathrm{d}t}. \tag{206}$$

Since $\overline{\epsilon} > 0$, this requires $\sigma_U < 0$, as expected for decay. Substituting (206) into (205) and dividing by $-2\overline{k}$,

$$\frac{\mathrm{d}\sigma_U}{\mathrm{d}t} = 2\,(C_{\epsilon 2} - 1)\,\sigma_U^2, \tag{207}$$

so $C_{\epsilon 2}$ follows at each instant from the scaling rate alone,

$$C_{\epsilon 2} = 1 + \frac{1}{2\sigma_U^2}\frac{\mathrm{d}\sigma_U}{\mathrm{d}t} = 1 - \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{\sigma_U}\right), \tag{208}$$

whether or not $C_{\epsilon 2}$ is constant. Under the power law, $1/\sigma_U$ is linear in t (194), so

$$C_{\epsilon 2} = 1 - \frac{\beta_U + 1}{2\beta_U} = \frac{\beta_U - 1}{2\beta_U} \tag{209}$$

is constant. Conversely, if $C_{\epsilon 2}$ is constant, (207) integrates to $1/\sigma_U = 1/\sigma_{U,0} - 2(C_{\epsilon 2} - 1)\,t$, which has the form of (194) with the decay exponent

$$\beta_U = -\frac{1}{2C_{\epsilon 2} - 1}. \tag{210}$$

Writing $\beta_U = -n/2$, so that $\overline{k^*} = U^2\,\overline{k} \propto (t^* + \alpha_U)^{-n}$, (209) is

$$C_{\epsilon 2} = \frac{n+2}{2n}, \tag{211}$$

not the classical $(n+1)/n$ [2].

The true constant. The constant of the true flow is the classical one. At $\nu_\mathrm{c}^* = 0$ the production and dissipation of ϵ_c^* in (173) are separately indeterminate, so $C_{\epsilon 2,\mathrm{c}}$ is defined by closing the whole rate of change of $\overline{\epsilon_\mathrm{c}^*}$,

$$\frac{\mathrm{d}\overline{\epsilon_\mathrm{c}^*}}{\mathrm{d}t^*} = -C_{\epsilon 2,\mathrm{c}}\,\frac{\left(\overline{\epsilon_\mathrm{c}^*}\right)^2}{\overline{k_\mathrm{c}^*}}. \tag{212}$$

Substituting (202), the factors $1 + r$ cancel and

$$C_{\epsilon 2,\mathrm{c}} = -\frac{\overline{k^*}}{\left(\overline{\epsilon^*}\right)^2}\,\frac{\mathrm{d}\overline{\epsilon^*}}{\mathrm{d}t^*}, \tag{213}$$

so $C_{\epsilon 2,\mathrm{c}}$ is computed from the simulated flow alone. The simulated flow obeys, in physical variables, the homogeneous form of (180),

$$\frac{\mathrm{d}\overline{\epsilon^*}}{\mathrm{d}t^*} = \frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*}\,\overline{\epsilon^*} - C_{\epsilon 2}\,\frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}}, \tag{214}$$

and substituting it into (213),

$$C_{\epsilon 2,\mathrm{c}} = C_{\epsilon 2} - \frac{\overline{k^*}}{\overline{\epsilon^*}}\,\frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*}\,: \tag{215}$$

the two constants differ by the eddy-viscosity change, which the simulated $C_{\epsilon 2}$ leaves out and the true $C_{\epsilon 2,\mathrm{c}}$ absorbs. With L constant, $\nu_\mathrm{t}^* = UL\,\nu_\mathrm{t}$ changes at the rate of U, and with $\mathrm{d}t^*/\mathrm{d}t = \tau = L/U$,

$$\frac{1}{\nu_\mathrm{t}^*}\frac{\mathrm{d}\nu_\mathrm{t}^*}{\mathrm{d}t^*} = \frac{1}{U}\frac{\mathrm{d}U}{\mathrm{d}t^*} = \frac{U}{L}\,\sigma_U, \qquad \frac{\overline{k^*}}{\overline{\epsilon^*}} = \frac{L}{U}\,\frac{\overline{k}}{\overline{\epsilon}} = -\frac{L}{U}\,\frac{1}{2\sigma_U}, \tag{216}$$

the last by (206). Their product is $-\frac{1}{2}$ at every instant, and (215) gives

$$C_{\epsilon 2,\mathrm{c}} = C_{\epsilon 2} + \frac{1}{2}, \tag{217}$$

the homogeneous case of (183) with $C_{\epsilon\nu} = \frac{1}{2}$. Within the spectral model this is exact: it uses neither the power law nor the smallness of r, only that the kinetic energy constraint holds $\overline{k}$ fixed. With the power law, (211) gives $C_{\epsilon 2,\mathrm{c}} = (n+2)/(2n) + \frac{1}{2} = (n+1)/n$, the classical result. The same value follows directly from (212): with $\overline{k_\mathrm{c}^*} \propto (t^* + \alpha_U)^{-n}$, $\overline{\epsilon_\mathrm{c}^*} = n\,\overline{k_\mathrm{c}^*}/(t^* + \alpha_U)$ and $\mathrm{d}\overline{\epsilon_\mathrm{c}^*}/\mathrm{d}t^* = -n(n+1)\,\overline{k_\mathrm{c}^*}/(t^* + \alpha_U)^2$, so $C_{\epsilon 2,\mathrm{c}} = (n+1)/n$.

In the self-similar frame the same relation appears term by term. By (206), the eddy-viscosity change of (204) is

$$\sigma_U\,\overline{\epsilon} = -\frac{1}{2}\,\frac{\overline{\epsilon}^{\,2}}{\overline{k}}, \tag{218}$$

of exactly the form of the closure: it is the homogeneous case of (181). Substituting it into (205),

$$\frac{\mathrm{d}\overline{\epsilon}}{\mathrm{d}t} = -3\sigma_U\,\overline{\epsilon} - \left(C_{\epsilon 2} + \frac{1}{2}\right)\frac{\overline{\epsilon}^{\,2}}{\overline{k}}, \tag{219}$$

which is the self-similar form of the constant-viscosity budget (212), with the constant $C_{\epsilon 2} + \frac{1}{2}$, and the homogeneous case of (185).

8 The turbulent puff

This section carries the comparison of §7, between the simulated flow and the true flow, over to the turbulent puff, using the budgets of §6 integrated over the puff. The pointwise budgets are reported in §9.

8.1 The simulated and the true flow

This subsection carries the comparison of §7, between the simulated flow and the true flow with $\nu_{\mathrm{c}}^* = 0$ (187), over to the turbulent puff decay case.

The comparison of §7.2 rests on three steps. (a) The two flows share a self-similar spectrum below the cutoff of the simulated flow. (b) Since $\nu_{\mathrm{t}}^* \propto UL$, the cutoff is fixed in the self-similar frame, so the energy of the tail that the simulation misses is a constant fraction r of $\overline{k}$ (201). To remove the effect of transport terms we will study volume integrated kinetic energy and dissipation instead of the instantaneous counter parts.

Notation and assumptions. The fields, budgets and assumptions are those of §6.1.4, in the stationary state of §6.1.3, with the budgets in the radial forms of §6.2 and §6.3. An integral without limits, $\int \cdot\,\mathrm{d}\mathbf{x}$, is over the sphere $r \le r_0$ in which the budgets hold. We assume that the puff has decayed before r_0, so that the profiles and their fluxes vanish on the surface of the sphere. The total resolved energy, which is then the mean of K_{res} of §5.7, is

$$K = \int \overline{k}\,\mathrm{d}\mathbf{x}, \qquad K^* = \int \overline{k^*}\,\mathrm{d}\mathbf{x}^* = U^2L^3\,K, \tag{220}$$

and by (164) the total dissipation is $\int \overline{\epsilon^*}\,\mathrm{d}\mathbf{x}^* = U^3L^2\int \overline{\epsilon}\,\mathrm{d}\mathbf{x}$. As in §7, the subscript c marks the true flow. In line with step (a), we assume that it shares the large scales of the simulated flow: it is self-similar with the same U and L, so that $\overline{k_{\mathrm{c}}^*} = U^2\,\overline{k_{\mathrm{c}}}$ and $\overline{\epsilon_{\mathrm{c}}^*} = (U^3/L)\,\overline{\epsilon_{\mathrm{c}}}$ with $\overline{k_{\mathrm{c}}}$ and $\overline{\epsilon_{\mathrm{c}}}$ functions of r alone. At $\nu_{\mathrm{c}}^* = 0$ the dissipation is indeterminate, and, as in (195), it is defined by the kinetic energy budget of the true flow, (167) without the viscous diffusion,

$$0 = \sigma_L\,r\frac{\mathrm{d}\overline{k_{\mathrm{c}}}}{\mathrm{d}r} - 2\sigma_U\,\overline{k_{\mathrm{c}}} + T_{k,\mathrm{c}} + \Pi_{k,\mathrm{c}} - \overline{\epsilon_{\mathrm{c}}}, \tag{221}$$

with $T_{k,\mathrm{c}}$ and $\Pi_{k,\mathrm{c}}$ the transport terms of the true flow.

8.2 Integrated budgets

The energy. Integrate the stationary budget (167) over the sphere. Every term is a function of r alone, so $\int \cdot\,\mathrm{d}\mathbf{x} = 4\pi\int_0^{r_0} \cdot\, r^2\,\mathrm{d}r$, and with K from (220)

$$0 = 4\pi\sigma_L\int_0^{r_0} r^3\frac{\mathrm{d}\overline{k}}{\mathrm{d}r}\,\mathrm{d}r - 2\sigma_U K + 4\pi\int_0^{r_0}(T_k + \Pi_k + D_k)\,r^2\,\mathrm{d}r - \int\overline{\epsilon}\,\mathrm{d}\mathbf{x}.$$

The transport terms T_k and Π_k are divergences, and so is the viscous diffusion $D_k = \partial_j(\nu_\mathrm{t}\,\partial_j\overline{k})$, since ν_t is uniform. In the radial forms (166), the factor r^2 of the volume element cancels the $1/r^2$ of each divergence, and the three reduce to fluxes through the surface of the sphere,

$$4\pi\int_0^{r_0}(T_k+\Pi_k+D_k)\,r^2\,\mathrm{d}r = 4\pi\left[r^2\left(-\overline{u_r k}-\overline{u_r p}+\nu_\mathrm{t}\frac{\mathrm{d}\overline{k}}{\mathrm{d}r}\right)\right]_0^{r_0}=0,$$

since the fluxes vanish at $r=r_0$ and the factor r^2 vanishes at $r=0$. The advection by the dilating frame is not a divergence. Integrating it by parts,

$$4\pi\int_0^{r_0} r^3\frac{\mathrm{d}\overline{k}}{\mathrm{d}r}\,\mathrm{d}r = 4\pi\left[r^3\,\overline{k}\right]_0^{r_0}-3\int_0^{r_0}4\pi r^2\,\overline{k}\,\mathrm{d}r=-3K,$$

since $\overline{k}(r_0)=0$. The integrated budget is therefore $0=-3\sigma_L K-2\sigma_U K-\int\overline{\epsilon}\,\mathrm{d}\mathbf{x}$, and with $\sigma_X=c_\tau\beta_X$ (156)

$$\int\overline{\epsilon}\,\mathrm{d}\mathbf{x} = -\left(2\sigma_U+3\sigma_L\right)K=-c_\tau\left(2\beta_U+3\beta_L\right)K=n\,c_\tau K,\qquad n=-\left(2\beta_U+3\beta_L\right). \tag{222}$$

The two rates are those of the scale factors of $K^*=U^2L^3K$ (220): $2\sigma_U$ that of U^2, which scales the energy per unit mass, and $3\sigma_L$ that of L^3, which scales the volume over which the puff spreads. Since K is constant, $\mathrm{d}K^*/\mathrm{d}t^*=(2\sigma_U+3\sigma_L)\,K^*/\tau$, so $K^*\propto(t^*+\alpha)^{-n}$ and n is the decay exponent of the total energy. Multiplied by $U^3L^2=U^2L^3/\tau$, (222) becomes $\int\overline{\epsilon^*}\,\mathrm{d}\mathbf{x}^*=-\mathrm{d}K^*/\mathrm{d}t^*$: the total energy decays at the rate of the total dissipation. It is the counterpart of (206), to which it reduces for uniform statistics and $\sigma_L=0$. Its two sides are measured independently, the rates from the scales and the integrals from the profiles, so it is also a check of the profiles.

The viscosity enters the derivation only through D_k, which integrates to zero, so the true flow obeys the same relation. Integrating (221) gives $\int\overline{\epsilon_\mathrm{c}}\,\mathrm{d}\mathbf{x}=n\,c_\tau K_\mathrm{c}$, with $K_\mathrm{c}=\int\overline{k_\mathrm{c}}\,\mathrm{d}\mathbf{x}$, and therefore

$$\frac{1}{K_\mathrm{c}}\int\overline{\epsilon_\mathrm{c}}\,\mathrm{d}\mathbf{x}=\frac{1}{K}\int\overline{\epsilon}\,\mathrm{d}\mathbf{x}=n\,c_\tau. \tag{223}$$

The ratio of the total dissipation to the total energy is a property of the large scales. Assuming the LES is well resolved so that $K_c\approx K$, it therefore follows that

$$\int\overline{\epsilon_c}\mathrm{d}\boldsymbol{x}\approx\int\overline{\epsilon}\mathrm{d}\boldsymbol{x}. \tag{224}$$

The constants. Integrate the dimensional budget (172) over the sphere $|\mathbf{x}^*|\le Lr_0$, which dilates with the frame. Since $\overline{\epsilon^*}$ vanishes on its surface, the time derivative may be taken outside the integral, and the divergences T^*_ϵ, Π^*_ϵ and D^*_ϵ integrate to zero:

$$\frac{\mathrm{d}}{\mathrm{d}t^*}\int\overline{\epsilon^*}\,\mathrm{d}\mathbf{x}^*=\frac{1}{\nu^*_\mathrm{t}}\frac{\mathrm{d}\nu^*_\mathrm{t}}{\mathrm{d}t^*}\int\overline{\epsilon^*}\,\mathrm{d}\mathbf{x}^*+\int\left(P^*_\epsilon-\Upsilon^*\right)\mathrm{d}\mathbf{x}^*. \tag{225}$$

Since K is constant, the total dissipation, $U^3L^2\int\overline{\epsilon}\,\mathrm{d}\mathbf{x}=n\,c_\tau\,U^3L^2K$ by (222), changes at the rate $(3\sigma_U+2\sigma_L)/\tau$ of U^3L^2 (4), and the eddy viscosity at the rate $(\sigma_U+\sigma_L)/\tau$ (174). The production and dissipation supply the difference,

$$\int\left(P^*_\epsilon-\Upsilon^*\right)\mathrm{d}\mathbf{x}^*=\frac{2\sigma_U+\sigma_L}{\tau}\int\overline{\epsilon^*}\,\mathrm{d}\mathbf{x}^*. \tag{226}$$

Closing the two terms on the right of (225) with (178) and (182), whose drift term integrates to a vanishing surface flux, gives

$$\frac{\mathrm{d}}{\mathrm{d}t^*}\int\overline{\epsilon^*}\,\mathrm{d}\mathbf{x}^*=-\Big(\underbrace{C_{\epsilon2}}_{P^*_\epsilon-\Upsilon^*}+\underbrace{C_{\epsilon\nu}}_{\text{eddy viscosity}}\Big)\int\frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}}\,\mathrm{d}\mathbf{x}^*. \tag{227}$$

Integrated, neither closure is an assumption: each equates a known integral to its constant times the integral on the right, and so defines the constant. That integral is $U^4L\int(\overline{\epsilon}^{\,2}/\overline{k})\,\mathrm{d}\mathbf{x}$, so dividing by $U^4L=U^3L^2/\tau$,

$$C_{\epsilon2}=-\left(2\sigma_U+\sigma_L\right)\frac{\int\overline{\epsilon}\,\mathrm{d}\mathbf{x}}{\int\left(\overline{\epsilon}^{\,2}/\overline{k}\right)\mathrm{d}\mathbf{x}},\qquad C_{\epsilon\nu}=-\left(\sigma_U+\sigma_L\right)\frac{\int\overline{\epsilon}\,\mathrm{d}\mathbf{x}}{\int\left(\overline{\epsilon}^{\,2}/\overline{k}\right)\mathrm{d}\mathbf{x}}.$$

By (222), the ratio of the integrals is

$$\frac{\int \overline{\epsilon}\,\mathrm{d}\mathbf{x}}{\int \left(\overline{\epsilon}^{\,2}/\overline{k}\right)\mathrm{d}\mathbf{x}} = \frac{\psi}{n\,c_\tau}, \qquad \psi = \frac{\left(\int \overline{\epsilon}\,\mathrm{d}\mathbf{x}\right)^2}{K\int \left(\overline{\epsilon}^{\,2}/\overline{k}\right)\mathrm{d}\mathbf{x}}, \tag{228}$$

with ψ a shape factor of the profiles, discussed below. With $\sigma_X = c_\tau \beta_X$ and $\beta_U = \beta_L - 1$, so that $n = 2 - 5\beta_L$,

$$C_{\epsilon 2} = \psi\,\frac{n+2-\beta_L}{2n}, \qquad C_{\epsilon\nu} = \psi\,\frac{n+\beta_L}{2n}, \qquad C_{\epsilon 2} + C_{\epsilon\nu} = \psi\,\frac{n+1}{n}. \tag{229}$$

The true flow, at $\nu_\mathrm{c}^* = 0$, has no eddy-viscosity. As in (212), its constant closes the whole rate of change of its dissipation, here with a drift transport as in (183), which again integrates to zero:

$$\frac{\mathrm{d}}{\mathrm{d}t^*}\int \overline{\epsilon_\mathrm{c}^*}\,\mathrm{d}\mathbf{x}^* = -C_{\epsilon 2,\mathrm{c}}\int \frac{\left(\overline{\epsilon_\mathrm{c}^*}\right)^2}{\overline{k_\mathrm{c}^*}}\,\mathrm{d}\mathbf{x}^*. \tag{230}$$

Compare this with the integrated budget (227) of the simulated flow. The two flows share U and L, so by (224) their total dissipations agree at every instant,

$$\int \overline{\epsilon_\mathrm{c}^*}\,\mathrm{d}\mathbf{x}^* = U^3L^2\int \overline{\epsilon_\mathrm{c}}\,\mathrm{d}\mathbf{x} \approx U^3L^2\int \overline{\epsilon}\,\mathrm{d}\mathbf{x} = \int \overline{\epsilon^*}\,\mathrm{d}\mathbf{x}^*,$$

and so do their rates of change, the left-hand sides of (227) and (230). Equating the right-hand sides,

$$C_{\epsilon 2,\mathrm{c}}\int \frac{\left(\overline{\epsilon_\mathrm{c}^*}\right)^2}{\overline{k_\mathrm{c}^*}}\,\mathrm{d}\mathbf{x}^* \approx (C_{\epsilon 2} + C_{\epsilon\nu})\int \frac{\left(\overline{\epsilon^*}\right)^2}{\overline{k^*}}\,\mathrm{d}\mathbf{x}^*. \tag{231}$$

The true constant is thus not $C_{\epsilon 2}$, the closure of $P_\epsilon^* - \Upsilon^*$ alone, but the sum of both closures of the simulated flow: the eddy-viscosity change, which the true flow lacks, is absorbed into its decay constant, the integrated counterpart of (215). The two integrals differ only through the shapes of the profiles. By (228), with ψ_c formed from the profiles of the true flow, they are U^4L times $(\int \overline{\epsilon_\mathrm{c}}\,\mathrm{d}\mathbf{x})^2/(\psi_\mathrm{c}K_\mathrm{c})$ and $(\int \overline{\epsilon}\,\mathrm{d}\mathbf{x})^2/(\psi K)$, so with $K_\mathrm{c} \approx K$, (224) and (229),

$$C_{\epsilon 2,\mathrm{c}} = \frac{\psi_\mathrm{c}}{\psi}\,(C_{\epsilon 2} + C_{\epsilon\nu}) = \psi_\mathrm{c}\,\frac{n+1}{n}. \tag{232}$$

If the true and simulated profiles have the same shapes, $\psi_\mathrm{c} = \psi$ and $C_{\epsilon 2,\mathrm{c}} = C_{\epsilon 2} + C_{\epsilon\nu}$, as in (183). $C_{\epsilon 2,\mathrm{c}}$ is ψ_c times the classical constant $(n+1)/n$ (below (217)), with n now the decay exponent of the total energy: since $3\sigma_U + 2\sigma_L = -(n+1)\,c_\tau$, the total dissipation decays as $(t^* + \alpha)^{-(n+1)}$, one power faster than the total energy.

Both equalities of (232) in fact hold exactly. Evaluating (230) directly, as (227) was, with (223) in place of (222), gives $C_{\epsilon 2,\mathrm{c}} = -\psi_\mathrm{c}(3\beta_U + 2\beta_L)/n = \psi_\mathrm{c}(n+1)/n$ with K_c cancelling. The result therefore needs neither $K_\mathrm{c} \approx K$ nor any closure of the simulated flow, only step (a): that the true flow is self-similar with the U and L of the simulated flow and vanishes at r_0.

To see what ψ measures, note that by (222) $n\,c_\tau$ is the $\overline{k}$-weighted mean of $\overline{\epsilon}/\overline{k}$ over the puff. Expanding the square,

$$\frac{1}{\psi} = 1 + \frac{1}{K}\int \left(\frac{\overline{\epsilon}}{n\,c_\tau\,\overline{k}} - 1\right)^2 \overline{k}\,\mathrm{d}\mathbf{x},$$

one plus the $\overline{k}$-weighted squared coefficient of variation of $\overline{\epsilon}/\overline{k}$. Hence $\psi \le 1$, with equality only for uniform $\overline{\epsilon}/\overline{k}$, and $1-\psi$ is of second order in its spread. The same holds for ψ_c by (223), so $\psi_\mathrm{c} - \psi$ is of second order in the spreads of $\overline{\epsilon}/\overline{k}$ and $\overline{\epsilon_\mathrm{c}}/\overline{k_\mathrm{c}}$.

With $\beta_L = 0$ and $\psi = 1$, (229) reduces to the homogeneous values (211) and $C_{\epsilon\nu} = \frac{1}{2}$ (217). In the puff the eddy viscosity changes with UL rather than with U, and the term $\psi\beta_L/(2n)$ moves part of the closure from $C_{\epsilon 2}$ to $C_{\epsilon\nu}$, leaving their sum unchanged. Since $n = 2 - 5\beta_L$, all the constants depend only on β_L, recovered as in §3.2, and on the shape factors ψ and ψ_c.

9 Navier–Stokes results

This section reports the pointwise budgets of the turbulent puff. The statistics, the transformation to the self-similar frame, and the exact and modeled budgets of $\overline{k}$ and $\overline{\epsilon}$ are those of §6; here we add the Reynolds stress budget and the kinetic energy budget at the origin.

9.1 Reynolds stress budget

Dimensional budget. Multiply (157) for u_i^* by u_j^*, add the same with i and j exchanged, and average. By continuity (77) the advection term is the divergence $\partial_k^*(u_i^* u_j^* u_k^*)$, and since ν_t^* is uniform the viscous term splits as

$$\nu_\mathrm{t}^* \left(u_j^* \frac{\partial^2 u_i^*}{\partial x_k^* \partial x_k^*} + u_i^* \frac{\partial^2 u_j^*}{\partial x_k^* \partial x_k^*} \right) = \nu_\mathrm{t}^* \frac{\partial^2 (u_i^* u_j^*)}{\partial x_k^* \partial x_k^*} - 2\nu_\mathrm{t}^* \frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}.$$

Unlike ϵ^*, the Reynolds stress $R_{ij}^* = \overline{u_i^* u_j^*}$ does not contain ν_t^*, so the time dependence of the eddy viscosity does not enter. The budget is [2]

$$\frac{\partial R_{ij}^*}{\partial t^*} = T_{ij}^* + \Pi_{ij}^* + D_{ij}^* - \epsilon_{ij}^*, \tag{233}$$

with

$$\begin{aligned}
T_{ij}^* &= -\frac{\partial}{\partial x_k^*} \overline{u_i^* u_j^* u_k^*} && \text{turbulent transport,} \\
\Pi_{ij}^* &= -\frac{1}{\rho} \left(\overline{u_i^* \frac{\partial p^*}{\partial x_j^*}} + \overline{u_j^* \frac{\partial p^*}{\partial x_i^*}} \right) && \text{velocity–pressure gradient,} \\
D_{ij}^* &= \nu_\mathrm{t}^* \frac{\partial^2 R_{ij}^*}{\partial x_k^* \partial x_k^*} && \text{viscous diffusion,} \\
\epsilon_{ij}^* &= 2\nu_\mathrm{t}^* \overline{\frac{\partial u_i^*}{\partial x_k^*} \frac{\partial u_j^*}{\partial x_k^*}} && \text{dissipation.}
\end{aligned} \tag{234}$$

The dissipation is the tensor (153) with $1/Re = 0$, so $\frac{1}{2}\epsilon_{ii}^* = \overline{\epsilon^*}$. The velocity–pressure gradient term contains both the pressure–strain redistribution and the pressure transport [2]; we keep it whole.

Self-similar frame. With $m_U = 2$, $m_L = 0$, (159) gives

$$\frac{\partial R_{ij}}{\partial t} = \sigma_L \, x_k \frac{\partial R_{ij}}{\partial x_k} - 2\sigma_U \, R_{ij} + T_{ij} + \Pi_{ij} + D_{ij} - \epsilon_{ij}, \tag{235}$$

with T_{ij}, Π_{ij}, D_{ij}, ϵ_{ij} of the form (234) without stars and with $\rho = 1$. The rescaling $-2\sigma_U R_{ij}$ is twice the velocity rescaling $-\sigma_U u_i$ of (99), since the stress is quadratic in the velocity.

Isotropic form. A symmetric second-rank tensor field invariant under rotations about the origin has the form

$$R_{ij} = R_\perp(r)\, \delta_{ij} + (R_r(r) - R_\perp(r))\, n_i n_j, \tag{236}$$

where $R_r = \overline{u_r u_r}$, with $u_r = n_i u_i$, is the radial normal stress and $R_\perp$ the normal stress in either direction tangent to the sphere. Every term of (235) has this form, so for a term X_{ij} we define its radial and tangential components

$$X_r = n_i n_j X_{ij}, \qquad X_\perp = \tfrac{1}{2}\,(\delta_{ij} - n_i n_j)\, X_{ij}, \tag{237}$$

evaluated as spherical means (149) of the projected pointwise integrands; the trace is $X_{ii} = X_r + 2X_\perp$. Since n_i is constant along rays, $x_k\, \partial_k n_i = 0$, so the advection term projects to $r\,\mathrm{d}R_r/\mathrm{d}r$ and $r\,\mathrm{d}R_\perp/\mathrm{d}r$. For the diffusion, the cross term $2\,\partial_k(R_r - R_\perp)\,\partial_k(n_i n_j)$ vanishes for the same reason, and $\nabla^2(n_i n_j) = (2\delta_{ij} - 6 n_i n_j)/r^2$, so

$$D_r = \nu_\mathrm{t} \left(\nabla_r^2 R_r - \frac{4}{r^2}\,(R_r - R_\perp) \right), \qquad D_\perp = \nu_\mathrm{t} \left(\nabla_r^2 R_\perp + \frac{2}{r^2}\,(R_r - R_\perp) \right). \tag{238}$$

The terms in $R_r - R_\perp$ couple the components through the curvature of the spheres and cancel in the trace. In the stationary state, (235) projects to

$$\begin{aligned} 0 &= \sigma_L\, r\frac{\mathrm{d}R_r}{\mathrm{d}r} - 2\sigma_U\, R_r + T_r + \Pi_r + D_r - \epsilon_r, \\ 0 &= \sigma_L\, r\frac{\mathrm{d}R_\perp}{\mathrm{d}r} - 2\sigma_U\, R_\perp + T_\perp + \Pi_\perp + D_\perp - \epsilon_\perp, \end{aligned} \tag{239}$$

each term a function of r alone. Regularity at the origin, where the stress is isotropic, requires $R_r(0) = R_\perp(0)$ and $R_r'(0) = R_\perp'(0) = 0$.

Half the trace of (233) is the kinetic energy budget (162): the pressure–strain part of Π^*_{ij} is traceless and only redistributes energy among the components, and $\frac{1}{2}\Pi^*_{ii} = \Pi^*_k$. Likewise, half the sum of the radial and twice the tangential equation of (239), with $\overline{k} = \frac{1}{2}(R_r + 2R_\perp)$ and $\overline{\epsilon} = \frac{1}{2}(\epsilon_r + 2\epsilon_\perp)$, is (167), which is therefore not an independent check of the profiles.

9.2 Kinetic energy budget at the origin

By (161) the advection term of (167) vanishes at $r = 0$, and

$$0 = -2\sigma_U\, \overline{k}(0) \underbrace{-3\frac{\mathrm{d}\overline{u_r k}}{\mathrm{d}r}\bigg|_0}_{T_k(0)} \underbrace{-3\frac{\mathrm{d}\overline{u_r p}}{\mathrm{d}r}\bigg|_0}_{\Pi_k(0)} + \underbrace{3\nu_\mathrm{t}\frac{\mathrm{d}^2\overline{k}}{\mathrm{d}r^2}\bigg|_0}_{D_k(0)} - \overline{\epsilon}(0). \tag{240}$$

The transport terms are divergences, so they integrate to zero over a volume on whose boundary the fluxes vanish, but locally they exchange energy between the centre and its surroundings: an outward turbulent flux near the origin gives $T_k(0) < 0$ and drains energy from the centre. The drift closure (168) reproduces this drain: at the origin its modeled transport is $-3C_{k\mathrm{T}}\,\sigma_L\,\overline{k}(0) < 0$ (§6.2.2).

A Navier–Stokes numerical details

This appendix collects two implementation details of the Navier–Stokes solver of §5: the construction of the initial condition and the evaluation of the scaling operator by spectral interpolation in three dimensions.

A.1 Initial condition

Because the curl of a vector feild is divergence-free we will define $\tilde{u}_i$ as the curl of some feild A_i

$$\tilde{u}_i = \varepsilon_{ijk}\frac{\partial A_k}{\partial x_j}, \qquad \text{equivalently} \qquad \hat{\tilde{u}}_i = \mathrm{i}\,\varepsilon_{ijk}\,\kappa_j \hat{A}_k. \tag{241}$$

$\tilde{u}_i$ is built on the grid (42) in four steps, followed by a rescaling.

1. *Noise.* Each component $A_k(\mathbf{x_n})$ is an independent standard normal, drawn from a seeded Mersenne Twister.

2. *Spectral filter.* The noise is given a spectrum,

$$\hat{A}_k \leftarrow \hat{A}_k\, e^{-\kappa^2/\kappa_p^2}. \tag{242}$$

3. *Window.* The potential is localized in physical space,

$$A_k \leftarrow A_k\, e^{-r^2/2\sigma^2}, \tag{243}$$

   with r the same distance from the centre of the domain that E_2 uses in (125).

4. *Rescaling.* Finally $\tilde{u}_i \leftarrow \tilde{u}_i\sqrt{E_0/F_1[\tilde{u}]}$, with F_1 evaluated as the Riemann sum (43), so that the field delivered to the integrator carries a prescribed E_0.

A.2 Spectral interpolation in three dimensions

The scaling operator (19) needs the same band-limited evaluation (27) as in one dimension, but at $d = 3$ a direct evaluation of it is no longer affordable, and its realization on the real transform differs enough from (73) to be worth writing out.

The sum on the real transform. Write $\hat{\tilde{u}}_i^{(k)}(\mathbf{j}) = N^{-3}\,\mathrm{RFFT}(\tilde{u}_k)_{\mathbf{j}}$ for the normalized coefficients, with the first axis carrying only the non-negative frequencies $j_1 = 0, 1, \ldots, N/2$ and the remaining two carrying the full signed set $j_d = 0, 1, \ldots, N/2, -(N/2-1), \ldots, -1$. Two corrections turn (27) into a sum over that layout. First, the omitted Hermitian conjugates are restored by the multiplicity

$$m_{j_1} = \begin{cases} 1, & j_1 = 0, \text{ or } j_1 = N/2 \text{ with } N \text{ even}, \\ 2, & \text{otherwise}, \end{cases} \tag{244}$$

exactly as the factor of 2 does in (73). Second, the transform phases its coefficients to the sample index n, which on the centred grid (42) places its origin at $x = -H/2$ rather than at $x = 0$. Since the scaling operator dilates about the centre of the domain, each mode must be evaluated at $\lambda_L x + H/2$, not at $\lambda_L(x + H/2)$ — the two agree only at $\lambda_L = 1$. As $H/2$ is half a period, the correction is the constant phase $\exp(\mathrm{i}\kappa_j H/2) = (-1)^j$ per axis. Together,

$$\Phi^{(\mathrm{S})}_{\Delta t}(\tilde{u}_k)_i(\mathbf{x}) = \frac{1}{\lambda_{U,k+1}}\,\mathrm{Re}\sum_{j_1=0}^{N/2}\sum_{j_2}\sum_{j_3} m_{j_1}\,\hat{\tilde{u}}_i^{(k)}(\mathbf{j})\prod_{d=1}^{3}(-1)^{j_d}\exp\!\left(\frac{2\pi\mathrm{i}\,j_d\,\lambda_{L,k+1}\,x_d}{H}\right). \tag{245}$$

Separable evaluation. Evaluated as written, (245) costs $O(N^{2d})$: every retained mode is summed at every grid point, which at $N = 128$ is 4×10^{12} complex multiply–adds per scaling substep. The kernel, however, is a product of one factor per axis, so the triple sum separates and can be applied one axis at a time. Define the two interpolation matrices

$$W^{(1)}_{nj_1} = m_{j_1}(-1)^{j_1}\exp\!\left(\frac{2\pi\mathrm{i}\,j_1\lambda_{L,k+1}x_n}{H}\right), \qquad W^{(\mathrm{f})}_{nj} = (-1)^j\exp\!\left(\frac{2\pi\mathrm{i}\,j\,\lambda_{L,k+1}x_n}{H}\right), \tag{246}$$

the first of size $N \times (N/2+1)$ for the half-spectrum axis and the second of size $N \times N$ for the two full axes, both rebuilt whenever $\lambda_{L,k+1}$ changes. Then (245) is the sequence of contractions

$$\hat{\tilde{u}}_i \xrightarrow{W^{(\mathrm{f})}\text{ on axis 3}} \xrightarrow{W^{(\mathrm{f})}\text{ on axis 2}} \xrightarrow{W^{(1)}\text{ on axis 1}} \mathrm{Re}(\cdot), \tag{247}$$

each of which is a matrix–matrix product once the contracted axis is brought to the front. The cost falls to $O(d\,N^{d+1})$, i.e. 8×10^8 multiply–adds at $N = 128$, a factor of 5×10^3 below the direct sum, and the work is expressed as three dense GEMM calls. The factorization is exact — it is a regrouping of the same finite sum, not an approximation — so the guarantee of §1.3 that the scaling step contributes no spatial discretization error is retained. For $d = 1$ the sequence (247) collapses to the single contraction with $W^{(1)}$, which is (73).

The Nyquist bin. For even N the bins $j_d = \pm N/2$ of a real field's transform coincide, and the two axis layouts resolve that differently. On the half-spectrum axis the mode is stored once and is given multiplicity $m_{N/2} = 1$ by (244). On a full axis it is stored in the single bin $j_d = N/2$ but represents the conjugate pair, so it must be split evenly between $+N/2$ and $-N/2$; the two halves combine to

$$W^{(\mathrm{f})}_{n,\,N/2} = (-1)^{N/2}\cos\!\left(\frac{\pi N\lambda_{L,k+1}x_n}{H}\right), \tag{248}$$

the same cosine that closes (73). With (248) in place the contraction (247) reproduces the direct sum (245) to roundoff for either parity of N.

The out-of-box cutoff. The box indicator χ of (28) is a product of one factor per axis, like the kernel, so it separates the same way. It is imposed by zeroing row n of each interpolation matrix (246) whenever the query of that axis leaves the box,

$$W^{(1)}_{nj_1} \leftarrow 0, \quad W^{(\mathrm{f})}_{nj} \leftarrow 0 \qquad \text{whenever } |x_n \lambda_{L,k+1}| > H/2. \tag{249}$$

After the three contractions (247) the point $\mathbf{x_n}$ then carries the product of the three row factors, which is $\chi(\mathbf{x_n}\lambda_{L,k+1})$ exactly. The cutoff is thus applied at no cost beyond building the matrices, and the direct sum it is tested against applies χ pointwise. For $\lambda_{L,k+1}$ within a few percent of unity the zeroed rows are the outermost one or two grid points per side. These lie deep in the sponge layer (§5.3), where the field has already been damped. Zeroing a region in physical space is not a solenoidal operation, so it reintroduces a divergence proportional to the amplitude it removes.

Loss of solenoidality and the Leray projection. The continuous scaling operator preserves continuity: if $\partial\tilde{u}_j/\partial x_j = 0$, then $\partial/\partial x_j\big[\lambda_U^{-1}\tilde{u}_j(\mathbf{x}\lambda_L)\big] = \lambda_U^{-1}\lambda_L\,(\partial\tilde{u}_j/\partial x_j)(\mathbf{x}\lambda_L) = 0$, the counterpart in physical space of (114). Its discrete evaluation does not preserve it exactly. The dilated field carries the wavenumbers $\lambda_L\boldsymbol{\kappa}$, which lie off the lattice of the grid, so sampling (245) at the grid points and transforming back redistributes each mode over neighbouring lattice wavenumbers without regard to $\kappa_i\hat{\tilde{u}}_i = 0$; the out-of-box cutoff (249) further removes amplitude in physical space. Both leave a divergence on the grid. When the option *project* of §2.5 is enabled, Scale therefore ends with the Leray projection of the scaled field back onto the solenoidal subspace, using the projection tensor (118),

$$\hat{\tilde{u}}_i(\boldsymbol{\kappa}) \leftarrow P_{il}(\boldsymbol{\kappa})\,\hat{\tilde{u}}_l(\boldsymbol{\kappa}) = \hat{\tilde{u}}_i - \frac{\kappa_i\kappa_l\hat{\tilde{u}}_l}{\kappa^2}, \qquad \boldsymbol{\kappa} \neq \mathbf{0}, \tag{250}$$

followed by the transform back to physical space. The mean mode $\boldsymbol{\kappa} = \mathbf{0}$ is left unchanged. The projection is orthogonal, so it can only remove energy, and the cost is one forward and one inverse transform per scaling step. Under Strang it is applied after each of the two scaling half steps, so the field entering Evolve is solenoidal as well as the field the step returns.

Interaction with dealiasing. The dilation maps each represented wavenumber to $\lambda_{L,k+1}\boldsymbol{\kappa}$, so a field that fills the spectrum up to the Nyquist wavenumber would wrap under any $\lambda_{L,k+1} > 1$. The 2/3 rule already required by the pseudo-spectral product (105) supplies the margin: the top retained mode sits at $N/3$ and moves to $\lambda_{L,k+1}N/3$, which stays below $N/2$ for every $\lambda_{L,k+1} < 3/2$. Runs in the self-similar frame hold $\lambda_{L,k+1}$ within a few percent of unity, so the margin is not approached, but it is the condition under which the interpolation stays free of aliasing. Finally, (247) is applied to each velocity component independently and mixes neither components nor the wavenumbers within an axis; at the exceptional value $\lambda_{L,k+1} = 1$ it reduces to the plain inverse transform, which is taken instead at $O(N^d \log N)$.

B Radial profiles from the spectral fields

This appendix sets out how the radial profiles of §6.1 are computed from a snapshot of the self-similar fields: the spherical mean (149) at a prescribed set of radii, the pressure that enters $\tilde{\Pi}_\epsilon$, the refined grid on which the integrands are formed, and the sampling in time.

B.1 The spherical mean of a band-limited field

Let $Q(\mathbf{x_n})$ be a real field on the grid (42), with discrete Fourier coefficients

$$\hat{Q}_\mathbf{m} = \sum_\mathbf{n} Q(\mathbf{x_n})\, e^{-\mathrm{i}\boldsymbol{\kappa}_\mathbf{m}\cdot\mathbf{n}\,\Delta x}, \qquad \boldsymbol{\kappa}_\mathbf{m} = \frac{2\pi}{H}\,\mathbf{m}, \tag{251}$$

over the signed indices $m_j \in \{-N/2, \ldots, N/2-1\}$. The transform is phased to the sample index, whose origin sits at $\mathbf{x} = -(H/2)(1,1,1)$ rather than at $\mathbf{x} = 0$, so its trigonometric interpolant is

$$Q(\mathbf{x}) = \frac{1}{N^3}\sum_\mathbf{m} \hat{s}_\mathbf{m}\, e^{\mathrm{i}\boldsymbol{\kappa}_\mathbf{m}\cdot\mathbf{x}}, \qquad \hat{s}_\mathbf{m} = (-1)^{m_1+m_2+m_3}\,\hat{Q}_\mathbf{m}, \tag{252}$$

with the phase-origin correction of (73) in each direction, since $\kappa_{m_j} H/2 = \pi m_j$. The spherical mean of a single mode is

$$\frac{1}{4\pi}\oint e^{\mathrm{i}\boldsymbol{\kappa}\cdot r\hat{\mathbf{n}}}\,\mathrm{d}\Omega = j_0(|\boldsymbol{\kappa}|\,r), \qquad j_0(z) = \frac{\sin z}{z}, \quad j_0(0) = 1, \tag{253}$$

so the spherical mean (149) of the interpolant is

$$\langle Q\rangle_r = \frac{1}{N^3}\sum_{\mathbf{m}} \mathrm{Re}\big(\hat{s}_{\mathbf{m}}\big)\, j_0(|\boldsymbol{\kappa}_{\mathbf{m}}|\,r), \tag{254}$$

the real part because Q is real and j_0 is even. Evaluated with the real transform, which stores the half spectrum $m_1 \geq 0$, each stored mode with $0 < m_1 < N/2$ stands also for its omitted conjugate and carries weight 2, as in (73); the planes $m_1 = 0$ and $m_1 = N/2$ carry weight 1. At $r = 0$, (254) is the interpolant at the origin, which is the grid value there.

Equation (254) is exact for the interpolant; no interpolation is done in physical space. For the resolved velocity, which is band-limited to the retained 2/3 band, the interpolant is the field itself. For a product such as ϵ, formed pointwise on the grid, it is the interpolant of the product as sampled, the same spectral interpolation as §1.3.

The kernel depends on $\boldsymbol{\kappa}_{\mathbf{m}}$ only through the integer $s = m_1^2 + m_2^2 + m_3^2 \leq 3(N/2)^2$. Summing the weighted coefficients over each such value first,

$$g_s = \sum_{\mathbf{m}:|\mathbf{m}|^2=s} c_{\mathbf{m}}\,\mathrm{Re}\big(\hat{s}_{\mathbf{m}}\big), \qquad \langle Q\rangle_r = \frac{1}{N^3}\sum_s g_s\, j_0\!\left(\frac{2\pi}{H}\sqrt{s}\,r\right), \tag{255}$$

with $c_{\mathbf{m}} \in \{1, 2\}$ the weights above, turns the sum into a regrouping of the spectrum, done once per field, followed by a product with the matrix $J_{is} = j_0\big((2\pi/H)\sqrt{s}\,r_i\big)$, tabulated once per run for the radii r_i below. The cost per field is one forward transform and a matrix–vector product of size $n_r \times 3(N/2)^2$.

The radial derivative of the profile follows from the same coefficients. Since $j_0' = -j_1$, with $j_1(z) = \sin z/z^2 - \cos z/z$,

$$r\frac{\partial\langle Q\rangle_r}{\partial r} = -\frac{1}{N^3}\sum_s g_s\,\kappa_s r\, j_1(\kappa_s r), \qquad \kappa_s = \frac{2\pi}{H}\sqrt{s}, \tag{256}$$

a second tabulated kernel applied to the same g_s. Because the spherical mean commutes with $x_j\,\partial/\partial x_j = r\,\partial/\partial r$, this is $\langle x_j\,\partial Q/\partial x_j\rangle_r$, the advection term of (175). It is evaluated this way rather than from the pointwise product $x_j\,\partial Q/\partial x_j$, which is not periodic and whose interpolant rings at the faces of the box. It vanishes at $r = 0$.

B.2 Radial projections of tensor fields

The Reynolds stress budget needs the spherical means of the projections (237) of symmetric tensor fields X_{ij}. The weight $n_i n_j = x_i x_j/r^2$ is not smooth at the origin, so $n_i n_j X_{ij}$ is not band-limited and the mean of its interpolant would not be exact. Instead, the projection is moved onto the kernel. Differentiating $\langle e^{\mathrm{i}\boldsymbol{\kappa}\cdot\mathbf{x}}\rangle_r = j_0(|\boldsymbol{\kappa}|r)$ twice with respect to $\boldsymbol{\kappa}$ gives

$$\frac{1}{4\pi}\oint n_i n_j\, e^{\mathrm{i}\boldsymbol{\kappa}\cdot r\hat{\mathbf{n}}}\,\mathrm{d}\Omega = \frac{j_1(z)}{z}\,\delta_{ij} - j_2(z)\,\hat{\kappa}_i\hat{\kappa}_j, \qquad z = |\boldsymbol{\kappa}|r, \tag{257}$$

with $\hat{\boldsymbol{\kappa}} = \boldsymbol{\kappa}/|\boldsymbol{\kappa}|$ and $j_2(z) = (3/z^2 - 1)\sin z/z - 3\cos z/z^2$. Hence $\langle n_i n_j X_{ij}\rangle_r$ is (255) applied to two fields of the components, the trace $\hat{X}_{ii}$ with the kernel $j_1(z)/z$ and the contraction $\hat{\kappa}_i\hat{\kappa}_j\hat{X}_{ij}$ with the kernel $-j_2(z)$. Both are formed in Fourier space from the band-limited components and grouped by s like any other field, so the projection is exact. The tangential projection follows as $X_\perp = \frac{1}{2}(\langle X_{ii}\rangle_r - X_r)$. The radial derivatives use $z\,\mathrm{d}(j_1/z)/\mathrm{d}z = -j_2$ and $z\,j_2' = zj_1 - 3j_2$, as in (256). At $r = 0$ the kernels are $1/3$ and 0, so $X_r = X_\perp = X_{ii}/3$.

B.3 The radii

The profiles are evaluated at $r = 0$ and at n_r logarithmically spaced radii,

$$r_i = r_{\min}\left(\frac{r_{\max}}{r_{\min}}\right)^{(i-1)/(n_r-1)}, \qquad i = 1, \ldots, n_r, \tag{258}$$

together with the few radii at which the statistical error is reported, which are added to the set exactly rather than taken from the nearest r_i. By default $n_r = 32$, $r_{\min} = \Delta x$, and $r_{\max} = r_0$, the radius inside which the sponge forcing vanishes and the ϵ budget of §6.3 applies.

B.4 The pressure

The pressure transport $\tilde{\Pi}_\epsilon$ needs the nondimensional pressure itself, which the projection (118) eliminates and the solver never forms. It is recovered from the Poisson equation (116) with the convective term in divergence form, $\hat{N}_l = \mathrm{i}\kappa_j \widehat{\tilde{u}_l \tilde{u}_j}$, which equals (105) for a solenoidal field,

$$\hat{\tilde{p}} = -\frac{\kappa_l \kappa_j \, \widehat{\tilde{u}_l \tilde{u}_j}}{\kappa^2}, \qquad \kappa \neq 0, \qquad \hat{\tilde{p}}(\mathbf{0}) = 0, \tag{259}$$

without the sponge term $\mathrm{i}\kappa_l \hat{F}_l/\kappa^2$ of (116), since the sponge is not part of the budget (§6.3). The products $\tilde{u}_l \tilde{u}_j$ are formed on the refined grid of §B.5, where they are exact. The solver itself forms the convective term in rotational form, $-\tilde{\mathbf{u}} \times \boldsymbol{\omega}$, which differs from $\tilde{u}_j \, \partial \tilde{u}_i / \partial x_j$ by the gradient $\partial_i(\frac{1}{2}\tilde{u}_k \tilde{u}_k)$; the pressure it implies is therefore the total head $\tilde{p} + \frac{1}{2}\tilde{u}_k \tilde{u}_k$, not the $\tilde{p}$ of the convective form in which the budget (175) is written. The subgrid term $\mathrm{i}\kappa_l \hat{G}_l/\kappa^2$ is zero under the self-similar constant eddy viscosity, since $\kappa_l \hat{G}_l = 0$. The scaling terms contribute nothing to the pressure (§5.5).

B.5 The integrands

The integrands of the ϵ budget terms (176) are cubic in $\tilde{u}_i$, and on the solver's grid they alias badly. The cubic mask leaves the field filled up to $|m_d| = N/3$ in every direction, so a product of n such fields reaches $|m_d| = nN/3$, and a cubic one reaches the box wavenumber N, twice the Nyquist wavenumber $N/2$. The aliased content folds onto the represented modes and corrupts both the interpolant between grid points, which the spherical mean (255) uses, and every spectral derivative of a product. The divergence terms T_ϵ and Π_ϵ suffer most.

We therefore form every product on a refined grid of $N_p = 2\lceil aN/2 \rceil$ points per direction, with padding factor $a \geq 1$, onto which the field is carried by zero-padding its spectrum. The normalized coefficients $\hat{Q}_{\mathbf{m}}/N^3$ of (251) carry over unchanged, so the field on the refined grid is the spectral interpolant of the stored one; the Nyquist modes of the solver's grid, which the mask keeps empty, are dropped. A product of n fields is then exact while $nN/3 < N_p/2$, that is for $a > 2n/3$: $a = 2$ makes every cubic integrand exact, and is the default. The eddy viscosity keeps the filter width of the run, $\Delta = H/N$, whatever N_p is, and $a = 1$ recovers evaluation on the solver's grid.

On the refined grid every derivative is taken spectrally, $\widehat{\partial Q/\partial x_j} = \mathrm{i}\kappa_j \hat{Q}$, and every product is formed pointwise in physical space. The divergence terms T_ϵ, Π_ϵ, and D_ϵ are formed as products first and differentiated spectrally afterwards, so that each is the divergence of the grid flux and not a sum of separately interpolated terms. Each integrand is then reduced to its profile by (255) on the refined grid, which is exact there as on the solver's grid.

B.6 Sampling in time

The profiles cost of order a hundred transforms per snapshot on the refined grid, against the handful of a physics step on the solver's grid, so they are not evaluated every step. They are evaluated on the stored field u_k every n_s steps, $n_s = \mathrm{round}(\Delta t_s/\Delta t)$, so that the samples are equally spaced in t at the interval Δt_s. The interval need only be short against the integral time scale of the profiles, which the batch means of Appendix C measure; by default $\Delta t_s = 0.02$, an order of magnitude below the integral time scale of $\epsilon(0)$ observed in the runs.

C Statistical error of time averages

This appendix sets out how the statistical error of a time-averaged quantity such as (150) is estimated, following [4]. Let $q(t)$ be a statistically stationary signal, here the spherical mean (149) at some radius r of the nondimensional dissipation rate, $q(t) = \langle\epsilon\rangle_r(t)$ of (155), or of the resolved kinetic energy, $q(t) = \langle\tilde{k}\rangle_r(t)$ of (122), both in the self-similar frame, with mean $\langle q\rangle$, fluctuation $q' = q - \langle q\rangle$, and root-mean-square fluctuation $q'_{\rm rms}$. Because the clock (83) is the advective time scale L/U, the self-similar time t is already measured in eddy turnover times.

C.1 Batch means

Successive values of q are correlated, so the samples of a run cannot be treated as independent. We instead use the method of batch means:

1. Discard the initial transient, during which the flow approaches the self-similar state, and let T be the length of the remaining record.

2. Split the record into $M = T/T_w$ non-overlapping windows of length T_w, and compute the window means
$$\langle q\rangle_i = \frac{1}{T_w}\int_{t_i}^{t_i+T_w} q(t)\,\mathrm{d}t, \qquad i = 1,\dots,M. \tag{260}$$
The samples $q_k = q(t_k)$ are recorded at equally spaced t_k, once per step (§2.3) or once per fixed sampling interval. Each integral (260) is evaluated as the exact integral of the piecewise-linear interpolant of the samples: the trapezoid rule on (t_k, q_k), with q interpolated linearly to window edges that fall between samples. The windows then tile $[t_0, t_0 + MT_w]$ exactly, and the remainder of the record shorter than T_w is dropped.

3. Estimate the mean and its standard error (SEM) as
$$q_{\rm est} = \frac{1}{M}\sum_{i=1}^{M}\langle q\rangle_i, \qquad \mathrm{SEM} \simeq \frac{\mathrm{STD}(\langle q\rangle_i)}{\sqrt{M}}. \tag{261}$$

4. Report the 95% confidence interval $q_{\rm est} \pm 1.96\,\mathrm{SEM}$.

The confidence interval assumes a Gaussian error, which the central limit theorem supplies when the windows are uncorrelated. Equation (261) holds only if

1. the windows are uncorrelated, $T_w \gg \tau_{\rm corr}$, where $\tau_{\rm corr}$ is the integral time scale of q; and

2. there are enough windows, $M \gg 1$.

For a fixed T the two conditions pull in opposite directions, so T_w has to be chosen with care.

C.2 Choosing the window length

The window length is chosen by plotting $\mathrm{SEM}\sqrt{T}$ against T_w, both in units of the eddy turnover time. The curve has three regimes:

- *Small T_w.* Adjacent windows are correlated, so M overcounts the independent samples and $\mathrm{SEM} \approx q'_{\rm rms}\sqrt{T_w/T}$, a slope of 1/2 on log axes. This underestimates the true error.

- *Plateau.* Both conditions hold and $\mathrm{SEM} \approx q'_{\rm rms}\sqrt{\tau_{\rm corr}/T}$, independent of T_w. This is the true SEM.

- *Large T_w.* M is too small and the estimate is noisy and unreliable.

The small-T_w asymptote and the plateau intersect at $T_w = \tau_{\mathrm{corr}}$, where

$$\tau_{\mathrm{corr}} = \frac{1}{q'^2_{\mathrm{rms}}} \int_{-\infty}^{\infty} C(\tau)\,\mathrm{d}\tau, \qquad C(\tau) = \langle q'(t)\, q'(t+\tau)\rangle, \tag{262}$$

so the plot also yields the integral time scale. A clear plateau requires a long record: [4] found one only for $T \gtrsim 10^4$ eddy turnover times, approximately 1.6×10^4 for homogeneous isotropic turbulence and 1.4×10^4 for channel flow at $Re_\tau = 180$. A dissipation-type quantity, being a second-order statistic of the velocity gradients, has τ_{corr} of roughly five eddy turnover times in homogeneous isotropic turbulence.

C.3 Pooling an ensemble of runs

A record long enough for a clear plateau can exceed the wall-clock limit of a single computing job. The record is then gathered as an ensemble of n independent runs instead: identical in every parameter except the seed of the random phases of the initial condition (§A.1), and in their length $T_{\mathrm{end},j}$. Each run is a draw from the same self-similar state, and after its own transient is discarded, run j contributes a record of length $T_j = T_{\mathrm{end},j} - t_{\mathrm{discard}}$.

The batch means of §C.1 extend to the ensemble without change. Every record is cut into windows of the common length T_w, starting at t_{discard}, giving $M_j = \lfloor T_j/T_w \rfloor$ window means $\langle q\rangle_{j,i}$ by (260), and the estimate (261) is formed over the union of all $M = \sum_j M_j$ windows,

$$q_{\mathrm{est}} = \frac{1}{M} \sum_{j=1}^{n} \sum_{i=1}^{M_j} \langle q\rangle_{j,i} = \sum_{j=1}^{n} \frac{M_j}{M}\, \bar{q}_j, \qquad \bar{q}_j = \frac{1}{M_j} \sum_{i=1}^{M_j} \langle q\rangle_{j,i}, \tag{263}$$

with SEM $\simeq$ STD$(\langle q\rangle_{j,i})/\sqrt{M}$ over the same union. Each run is therefore weighted by its length. When every window has the same variance, which holds for T_w on the plateau and runs sampling a single stationary state, this is the minimum-variance weighting of the run means $\bar{q}_j$, whose variances are inversely proportional to M_j. The two conditions of §C.1 are, if anything, better met than within a single run: windows of different runs are uncorrelated exactly, so only the windows adjacent within each run need $T_w \gg \tau_{\mathrm{corr}}$, and M grows with every run added. The window length is chosen as in §C.2, from the curve of SEM$\sqrt{T}$ against T_w for the pooled windows, with $T = \sum_j T_j$; the longest window scanned is the shortest T_j, so that every run still contributes. The cost of pooling is one transient t_{discard} discarded per run, rather than one in all.

Two runs that differ only in T_{end} follow the same trajectory, up to the roundoff that eventually separates them, and would be counted twice. For each seed, only the longest completed run is therefore used.

Pooling assumes that the runs sample the same state, and the ensemble can test it. Within run j the window means scatter about $\bar{q}_j$ with the within-run variance

$$s_w^2 = \frac{1}{M-n} \sum_{j=1}^{n} \sum_{i=1}^{M_j} \left(\langle q\rangle_{j,i} - \bar{q}_j\right)^2, \tag{264}$$

and if the run means differ only by that scatter, the ratio of the between-run to the within-run variance,

$$F = \frac{1}{(n-1)\, s_w^2} \sum_{j=1}^{n} M_j \left(\bar{q}_j - q_{\mathrm{est}}\right)^2, \tag{265}$$

follows the F distribution with $(n-1, M-n)$ degrees of freedom: close to one, with standard deviation $\approx \sqrt{2/(n-1)}$ for $M \gg n$ (the one-way analysis of variance). A value well above one means the runs disagree by more than their own fluctuations allow: the transient is not over by t_{discard}, the initial condition leaves a lasting imprint, or T_w is short of the plateau, which shrinks s_w. The unweighted mean of the $\bar{q}_j$, with its standard error STD$(\bar{q}_j)/\sqrt{n}$ from the spread of the runs alone, is reported alongside (263) as a second check that does not rely on the windows being uncorrelated.

The same pooling applies radius by radius to the spherical means of Appendix B, whose radii are shared by every run, and to the coefficients $C_{\epsilon 2,\mathrm{c}}$ and $C_{\epsilon \mathrm{T},\mathrm{c}}$, with their parts, of the modeled ϵ budget (§6.3.2), which are fitted to the pooled time averages and, for their confidence intervals, to the pooled windows.

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